

STEAMROOT · BUILD PLAN

SteamRoot prototype build plan

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	First build plan, with pictures by component and step; design made constructable (STR-DDR-003)

SteamRoot prototype build plan

Plan, not yet built. How to build the first proof-of-concept prototype, component by component. Building and testing to it is TRL 4 work. Decisions still to be made are kept in the design decisions register ([docs/06-design-decisions.md](#)), not here.

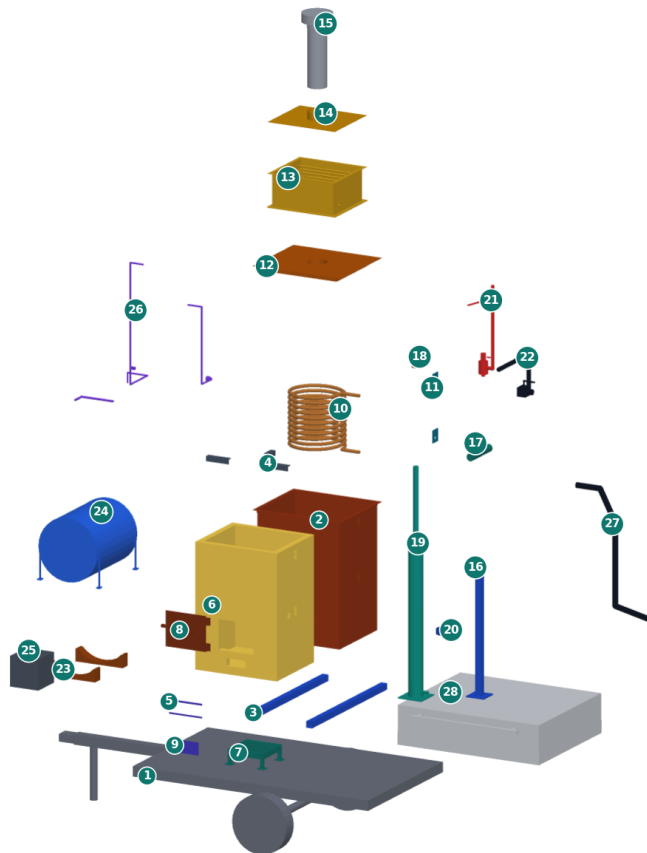
Safety: Nothing in this plan may be bought, built or fired until there is a written ruling from the local boiler authority on this open-vented, wood-fired steam coil (safety stop S1). SteamRoot combines an open fire, carbon monoxide and steam that burns skin in under a second. The steam side must always be open to the air through the water-seal vent: never fit a valve, cap or plug that could shut it. Welding, grinding and cutting ceramic fibre each have their own hazards, given in section 6.

1. What you are building

SteamRoot prototype: every component, pulled apart

Numbered in build order. Seen from the operator side (front), right and above; the second hood is the same as the first

- 1 Trailer, used (with wheels)
- 2 Firebox shell
- 3 Firebox skids (2)
- 4 Coil brackets (3)
- 5 Damper guides (2)
- 6 Fibre lining, walls and floor
- 7 Grate stand and cast grate
- 8 Door, hinges and latch
- 9 Air damper slide
- 10 Monotube coil
- 11 Tube gland plates (2)
- 12 Roof and roof board
- 13 Economizer box and tube bank
- 14 Economizer lid
- 15 Chimney and spark arrestor
- 16 Header post
- 17 Steam header and gauge
- 18 Coil outlet pipe
- 19 Water-seal pot and vent
- 20 Pot stay
- 21 Relief valve and discharge pipe
- 22 Diverter and vent line
- 23 Drum saddles (2)
- 24 Feed drum and straps
- 25 Pump and alarm box
- 26 Feed lines and jumper
- 27 Steam hose
- 28 Steam hood (2; one shown)



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Figure 1. Every component pulled apart and numbered in build order; the second hood is the same as the first.

The prototype is a small wood-fired steam generator on a used single-axle trailer, with two insulated steam hoods that are pressed onto a soil bed. On the trailer stand a lined steel firebox with a stainless steel coil above the fire, a flue-gas economizer and chimney on the firebox roof, a steam header on a post with its water-seal pot, vent, relief valve and diverter, a feed water drum on two saddles, and a box holding the feed pump, battery and alarms. A 6 m steam hose takes the steam to whichever hood is in use. Figure 1 shows the 28 components in the order you make or fit them. Most are made in a farm or small fabrication workshop: the firebox shell, roof, skids, brackets, grate stand, door and damper are cut and welded from mild steel sheet, tube and angle; the coil is wound from one length of stainless tube on a former; the economizer box and its tube bank, the header, seal pot and post are made from sheet, pipe and tube; the hoods are folded and riveted from aluminium and galvanised sheet. Everything else is bought: the trailer, the ceramic fibre lining, the cast grate, the chimney and spark arrestor, the certified relief valve, the diverter, the hose, the drum, the pump and the alarms. The parts cost about \$2,275 from the bill of materials.

■ **2. What changed to make it buildable**

The concept showed what SteamRoot does; some of its parts could not be made, fixed or put together as drawn. Each change below keeps what the machine does, and all of them are recorded in decision record STR-DDR-003, open for Amish's review.

Table 1. Changes from the concept.

Component	The concept had	The buildable design has	Why
Firebox and coil	A 650 mm firebox with the coil wound round the fire from near the floor to the roof; nowhere to load wood	A 1000 mm firebox with the door, grate and air inlet below the coil and 280 mm of fire space (Figure 9)	Wood has to go in through a door onto a grate the coil does not surround
Coil support	None	Three stainless brackets welded to the shell through the lining (Figure 6)	The coil rests on them and can grow when hot
Coil fitting	A welded box the coil could not enter; tails through the wall with no holes, crossing outside	A bolted roof; the coil goes in from the top and its tails slide out through slots closed by gland plates; unions outside (Figure 15)	The coil can be fitted, removed and allowed to grow
Economizer	A solid box, not fixed, with the coil inlet ending beside it	A bolted box on the roof with a serpentine tube bank inside, a lid carrying the chimney, and unions to the feed line and coil (Figure 18)	The feed water has a path from the pump to the coil
Firebox fixing	Standing loose on the deck	Two skids welded under the firebox, bolted into the trailer's side rails (Figure 4)	Held for towing, with an air gap under the hot floor
Water-seal pot	Reached through the deck; a 1 m dip leg that would have let the header reach about 0.11 bar	A pot on a foot plate with an 890 mm dip leg, so the water it pushes up outside adds the rest of the 1 m seal (Figure 25)	Keeps the 0.094 bar seal limit exactly and keeps water out of the vent
Header	Hung on the coil tube	A post with a saddle and U-bolts, and a stay to the pot (Figure 22)	The coil is not a structure
Diverter, relief valve	No pipe from the diverter to the vent; relief valve with no discharge pipe	A vent line from the diverter into the vent pipe; a relief discharge pipe to 2.3 m (Figure 22)	Steam always has a safe way out
Hose and hood	The hose passed through the deck and the hood; the manifold floated	A coupling on top of each hood, a riser to the hung manifold, a riveted skirt (Figure 32)	Every joint is made, not drawn through
Drum and feed	A cradle cut through the drum; no feed lines	Two hardwood saddles, ratchet straps, a suction hose and a stainless feed line (Figure 28)	The drum is held for towing and the water reaches the coil

■ 3. Making the components

Make and check each component before the assembly step that needs it. Sizes are in millimetres. "Front" is the operator side, where the firebox door is; "left" and "right" are as seen standing at the front facing the machine, so the drawbar is on the left and the hoods on the right. Heights are above the ground with the trailer level unless a step says otherwise. Workshop tolerance is 1 mm on steelwork and 0.5 mm on drilled holes unless a step says otherwise; drawings do not carry tolerances before TRL 4.

3.1 Trailer (bought, used)

What it is. A used single-axle garden or utility trailer with a flat deck about 2.0 x 1.2 m, about 500 above the ground, with steel side rails about 80 deep, rated for 750 kg or more.

What to do to it. Clean and check the deck, the side rails, the axle, the tyres, the lights and the coupling, and repair anything worn. Mark the deck centre line. Drill the holes for the skids (section 3.3), the header post and pot foot plates (sections 3.15 and 3.17), the drum saddles (section 3.19) and the pump box (section 3.23) once those parts are made, by marking through them in place.

Check before moving on. The trailer stands level on its jack; the axle rating on its plate covers 602 kg.

3.2 Firebox shell

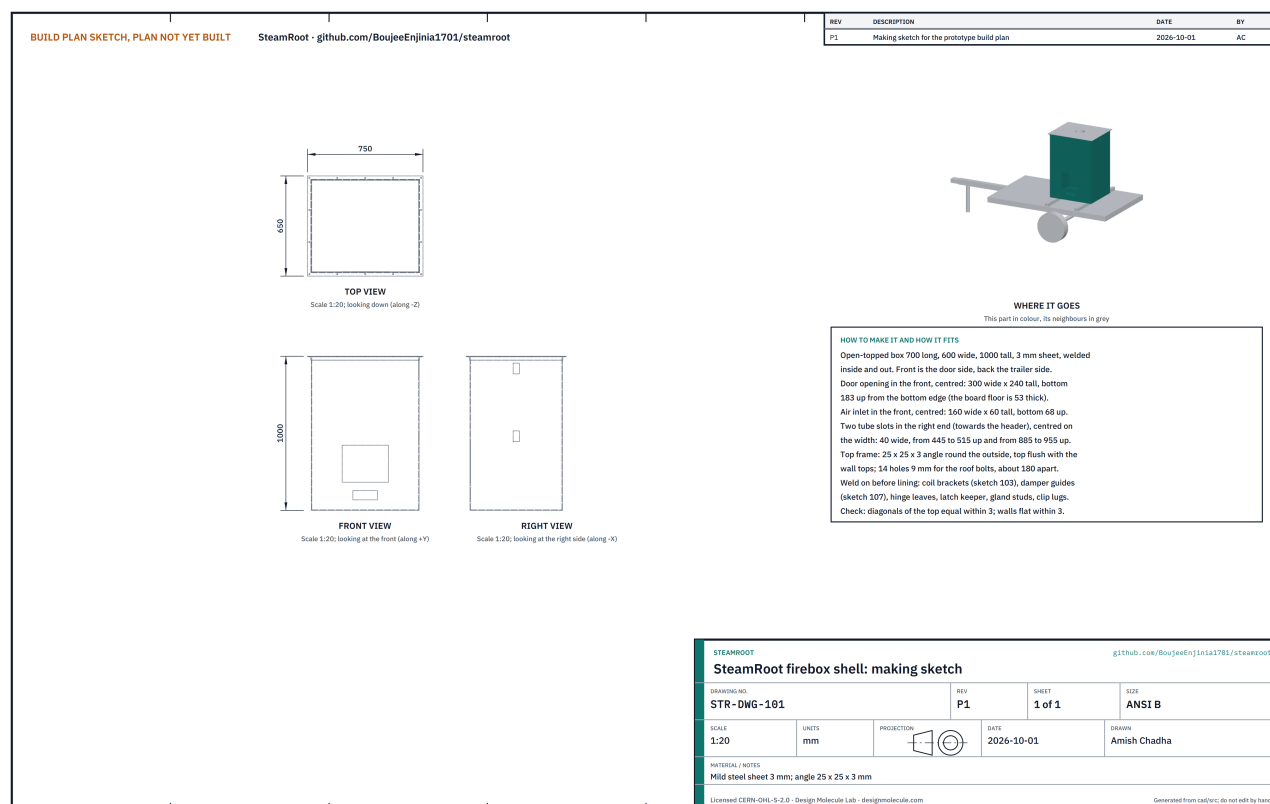


Figure 2. Firebox shell making sketch (STR-DWG-101).

What it is and what it is made from. The steel box that holds the lining, the grate and the coil. Mild steel sheet 3 mm, welded, open at the top, 700 long, 600 wide and 1000 tall, with a frame of 25 x 25 x 3 mm angle round the top edge for the roof bolts.

How to make it.

1. Cut the floor 700 x 600, the front and back 700 x 997 and the two ends 594 x 997 from 3 mm sheet. Mark the inside faces.
2. In the front, cut the door opening 300 wide x 240 tall, centred, its bottom 183 above the bottom edge of the box, and the air inlet 160 wide x 60 tall, centred, its bottom 68 above. Round the corners to about 5 and file the edges.
3. In the right end, cut the two tube slots 40 wide, centred on the width: one from 445 to 515 above the bottom edge, the other from 885 to 955.

4. Tack the walls to the floor and to each other, check the box is square (diagonals of the top within 3), then weld all seams inside and out.
5. Weld the 25 x 25 x 3 angle frame round the outside of the top edge, mitred at the corners, its top flush with the wall tops. Drill 14 holes of 9 mm through the frame, 12.5 out from the walls: five along each long side and two more at each end, evenly spaced.
6. Weld on, before the lining goes in: the coil brackets inside (section 3.4); the damper guides, the hinge leaves and the latch keeper outside on the front (sections 3.7 and 3.8); four M6 studs round each tube slot (section 3.10); two clip lugs on the left end for the feed line, 190 behind the centre line, 400 and 850 above the bottom edge.
7. Fit stainless lining pins on the inside walls about 250 apart (section 3.5). Paint the outside with high-temperature paint.

How it fits the parts next to it. The floor sits on two skids (Figure 4). The roof bolts down onto the angle frame (section 3.11). The economizer stands on the roof, not on the shell.

Check before moving on. The walls are flat within 3; the door opening and slots are where Figure 2 shows; a 40 mm bar passes freely through each slot.

3.3 Firebox skids (make 2)

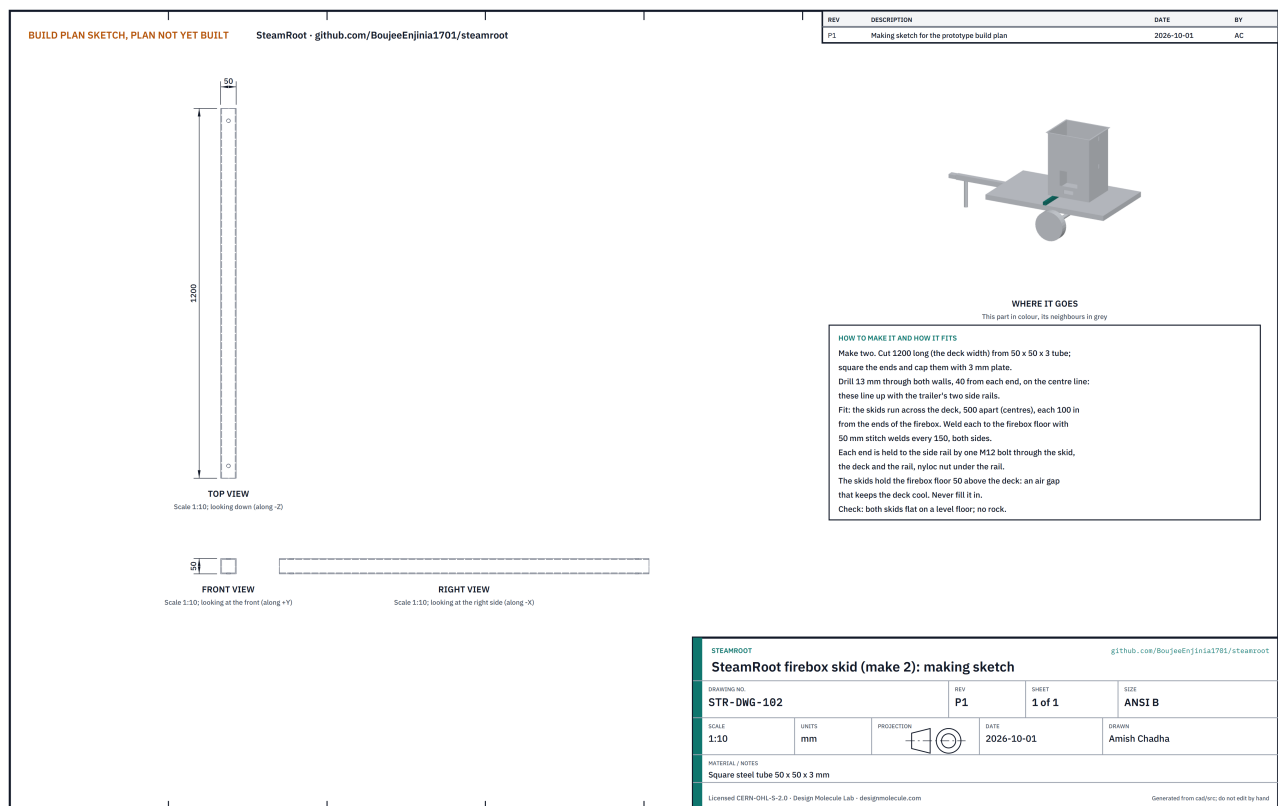


Figure 3. Firebox skid making sketch (STR-DWG-102).

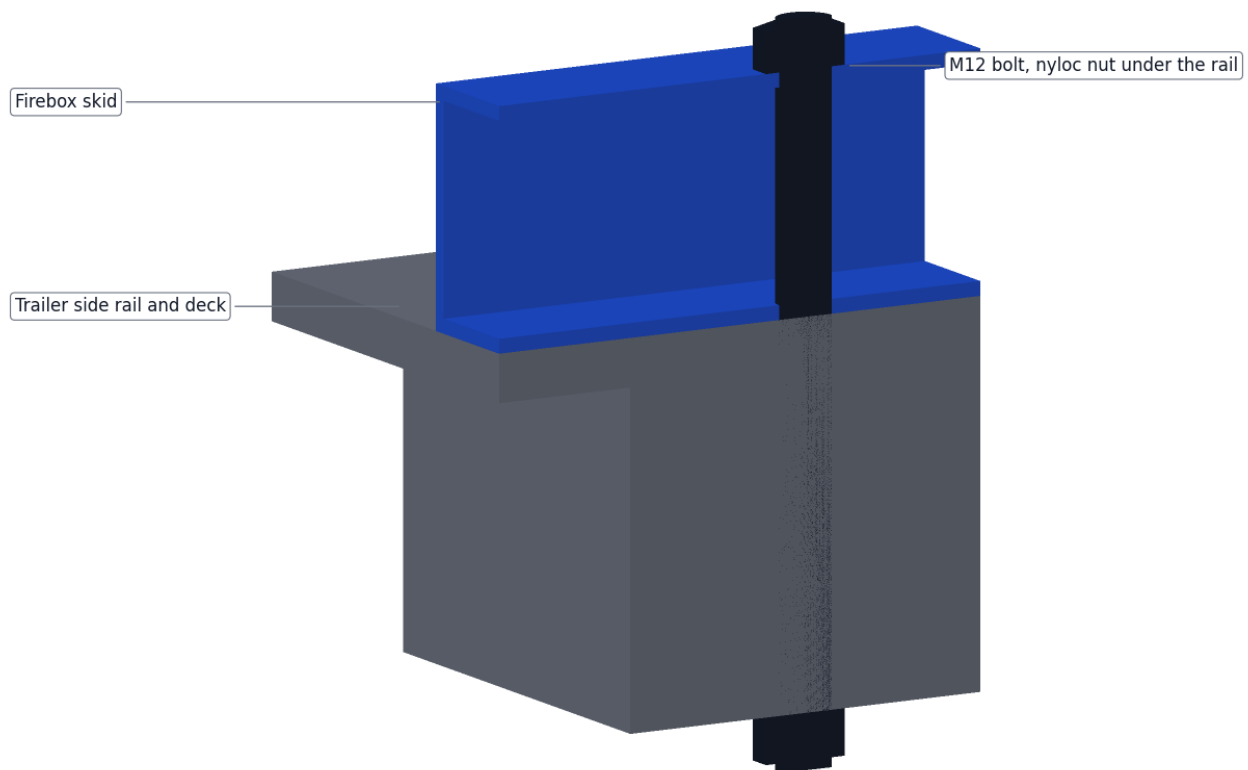
What it is and what it is made from. Two lengths of square tube that run across the deck under the firebox; their ends bolt into the trailer's side rails. Square steel tube 50 x 50 x 3 mm.

How to make it.

1. Cut two lengths of 1200 (the deck width); square the ends and weld a 3 mm cap on each.
2. Drill 13 mm through both walls, 40 from each end, on the centre line.
3. Turn the firebox over. Lay the skids across its floor, 500 apart (centres), each 100 in from an end, overhanging equally front and back. Stitch weld each to the floor, 50 long every 150, on both sides.

How it fits the parts next to it.**Joint 1: skid end on the trailer's side rail (back left corner)**

Cut through the bolt. The skid is welded under the firebox and bolted through the deck into the rail



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Figure 4. Each skid end is held by one M12 bolt through the skid, the deck and the side rail, with a nyloc nut under the rail.

The skids hold the firebox floor 50 above the deck. That air gap keeps the deck cool: never fill it.

Check before moving on. The firebox stands on a level floor on its skids without rocking.

3.4 Coil brackets (make 3)

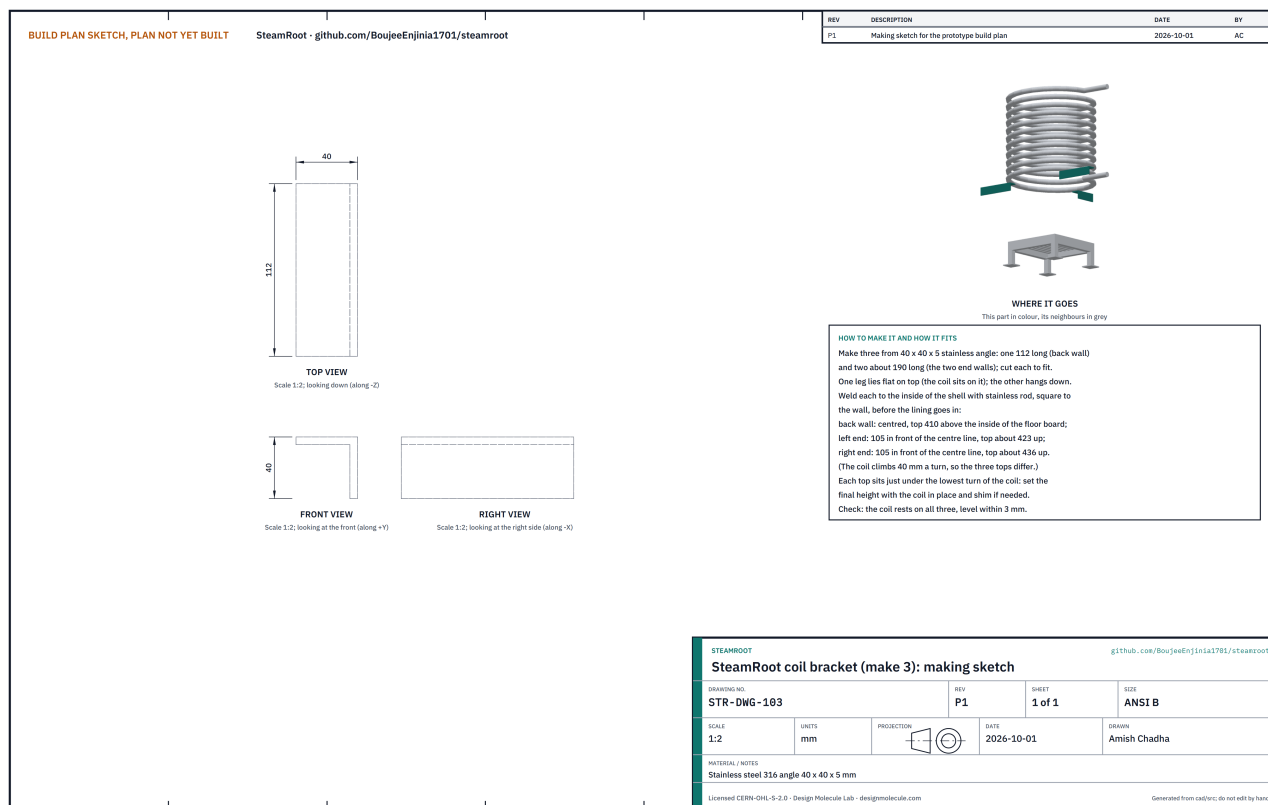


Figure 5. Coil bracket making sketch (STR-DWG-103).

What it is and what it is made from. Three short angles welded inside the firebox that carry the lowest turn of the coil. Stainless steel 316 angle 40 x 40 x 5 mm.

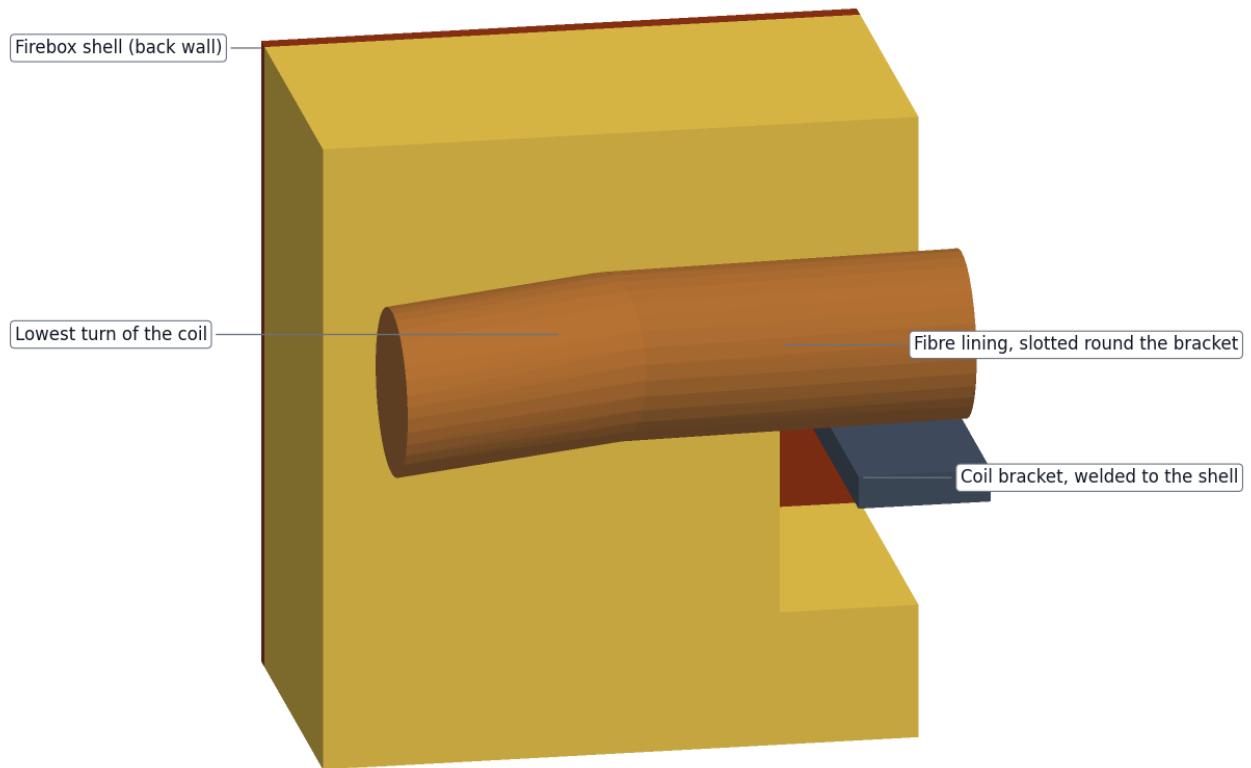
How to make it.

1. Cut one 112 long and two about 190 long; deburr.
2. Weld each to the inside of the shell with stainless rod (309 type for stainless to mild steel), the flat leg on top and the other leg hanging down: the short one in the middle of the back wall, its top 410 above the inside of the floor board (463 above the bottom of the shell); one on each end wall, 105 in front of the centre line, tops about 423 (left) and 436 (right) above the floor board. The coil climbs 40 mm a turn, so the three tops differ.
3. Set the final heights with the coil in place (step 7) and shim under the coil if needed.

How it fits the parts next to it.

Joint 2: coil bracket through the lining (back wall)

Cut open, seen from inside the firebox. The coil rests on the bracket; nothing else holds it



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Figure 6. The bracket passes through a slot in the lining; the lowest turn of the coil rests on it. Nothing else holds the coil.

Check before moving on. All three brackets are level across and square to their walls.

3.5 Fibre lining (bought, cut to fit)

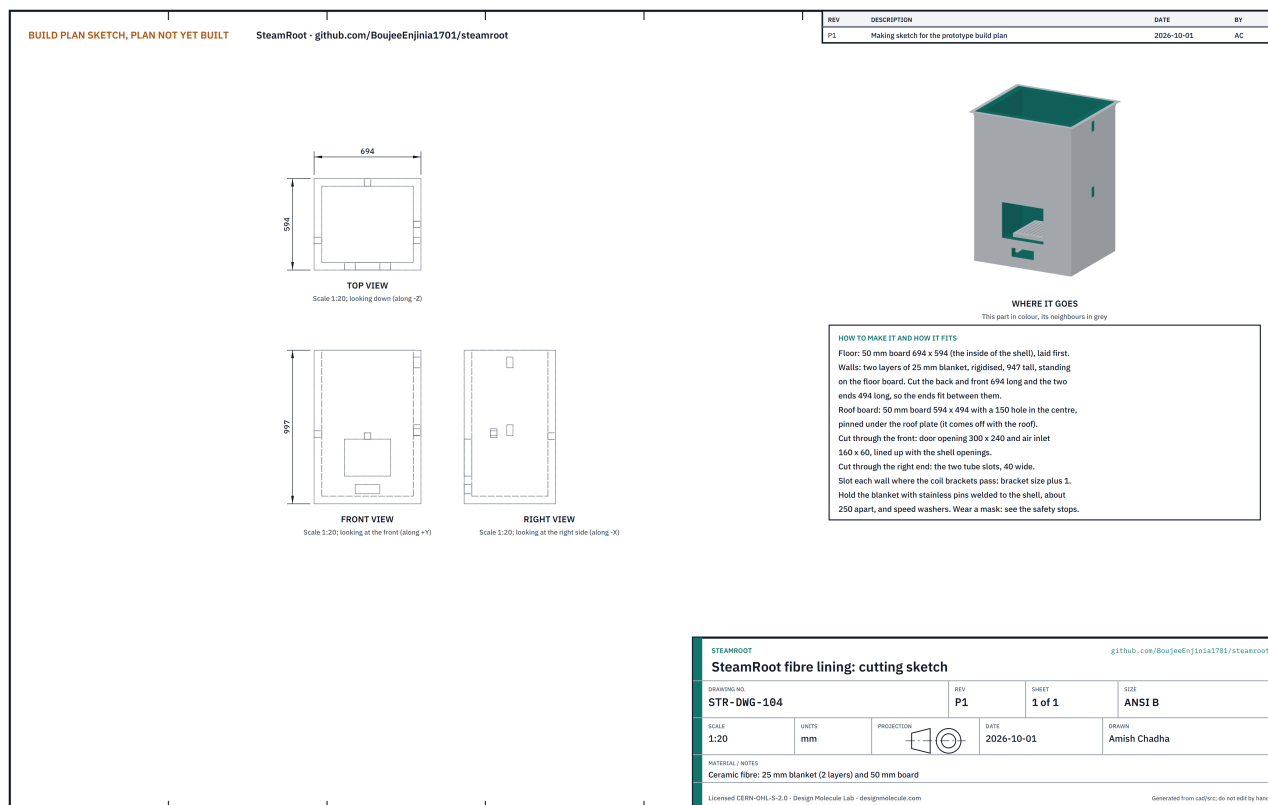


Figure 7. Fibre lining cutting sketch (STR-DWG-104).

What it is and what it is made from. The insulation inside the firebox: 50 mm ceramic fibre board on the floor and under the roof, and two layers of 25 mm rigidised ceramic fibre blanket on the walls.

How to make it.

1. Cut the floor board 694 x 594 and lay it in first.
2. Cut each layer of wall blanket 947 tall: the back and front 694 long, the two ends 494 long so they fit between them. Cut through the front for the door opening (300 x 240) and air inlet (160 x 60), and through the right end for the two tube slots, each lined up with the shell. Slot the blanket where the brackets pass (bracket size plus 1).
3. Press each layer over the pins, the joints of the second layer staggered from the first, and hold it with speed washers. Brush on rigidizer.
4. Cut the roof board 594 x 494 with a 150 hole in the centre; it goes on the roof (section 3.11).

How it fits the parts next to it. The walls stand on the floor board; the roof board fits inside the tops of the walls (Figure 18).

Check before moving on. No steel shows inside except the bracket tops and the pins; the lining is 50 thick everywhere.

3.6 Grate stand and cast grate

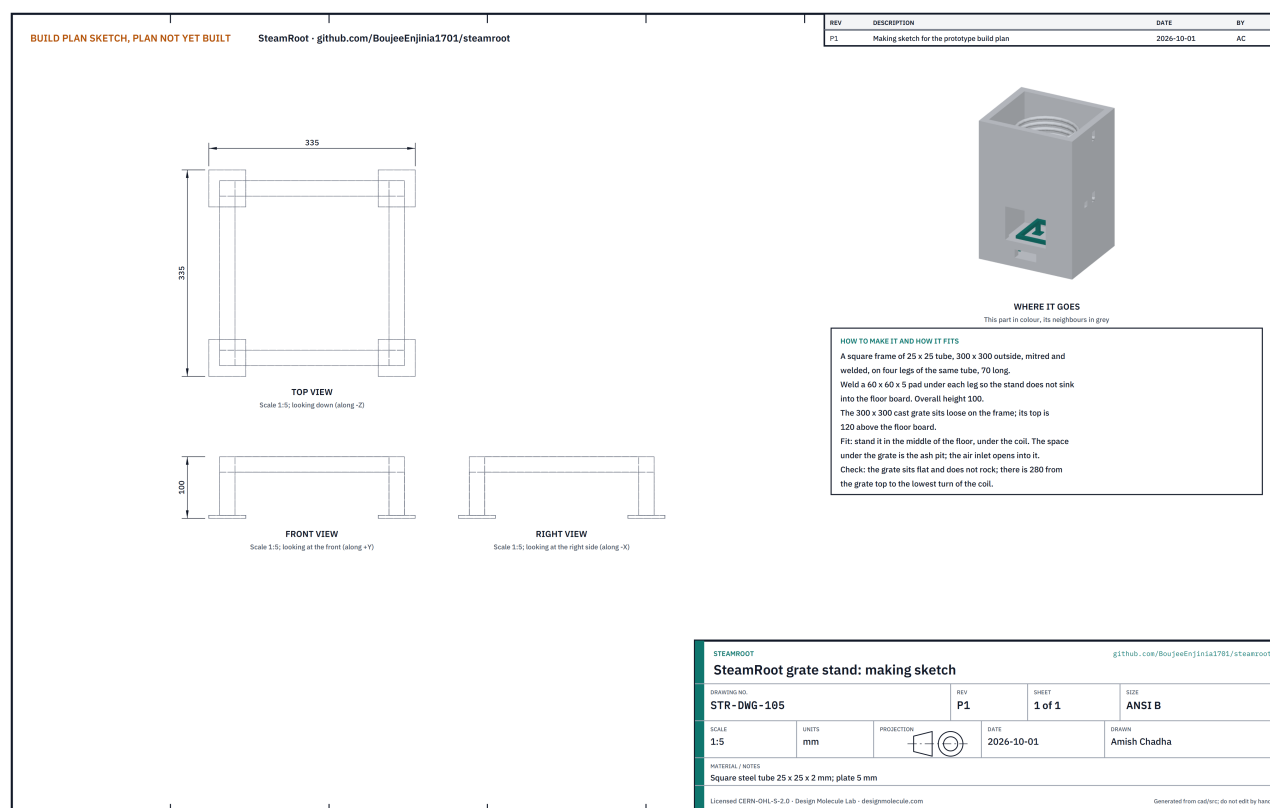


Figure 8. Grate stand making sketch (STR-DWG-105).

What it is and what it is made from. A small table that holds the cast grate 100 above the floor, so air comes in under the fire. Square steel tube 25 x 25 x 2 mm and 5 mm plate; a bought 300 x 300 cast iron grate.

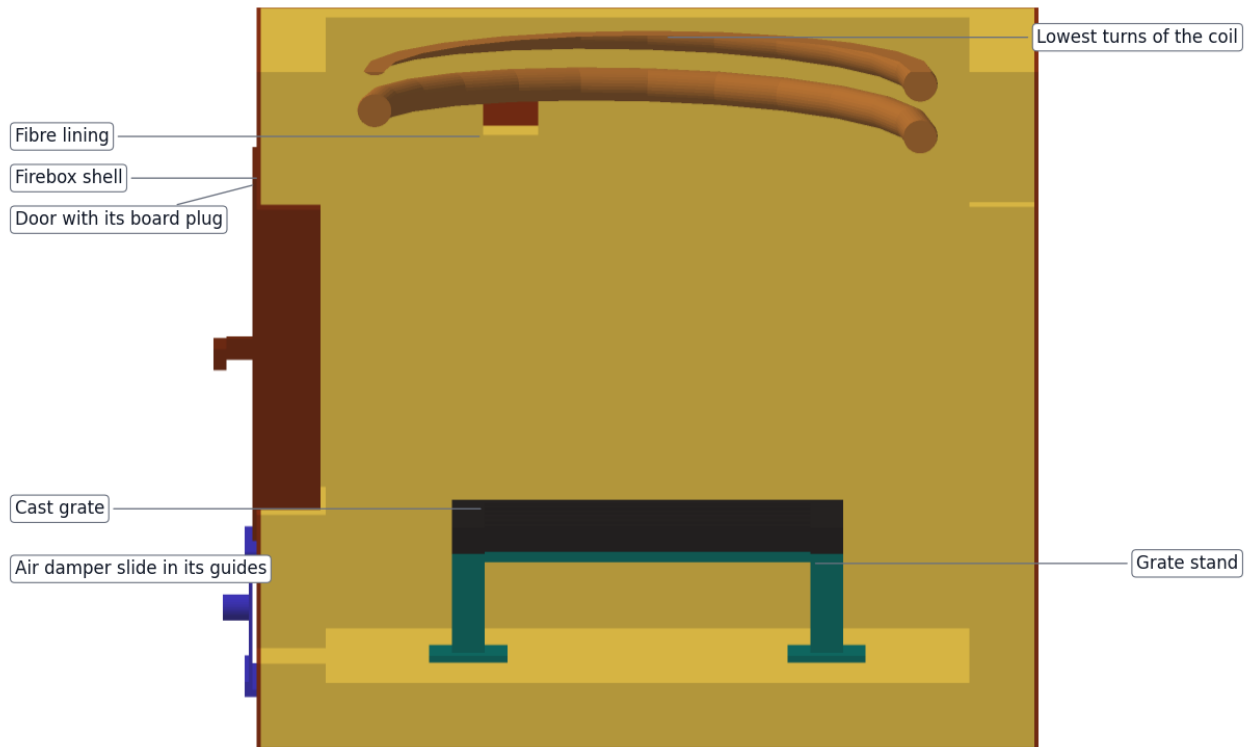
How to make it.

1. Cut four lengths of 300 with mitred ends and weld them into a square frame 300 x 300 outside.
2. Cut four legs 70 long and weld one under each corner; weld a 60 x 60 x 5 pad under each leg.
3. Check the grate sits flat on the frame.

How it fits the parts next to it.

Joint 3: the lower firebox, cut through the middle

Seen from the right. Air comes in under the grate; wood goes in through the door; the coil starts 280 above the grate



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Figure 9. The stand sits on the floor board in the middle of the firebox, under the coil. The air inlet opens into the ash pit under the grate; the door opening is level with the fire; the coil starts 280 above the grate.

Check before moving on. The grate sits flat and does not rock; there is 280 from the top of the grate to the lowest turn of the coil.

3.7 Firebox door

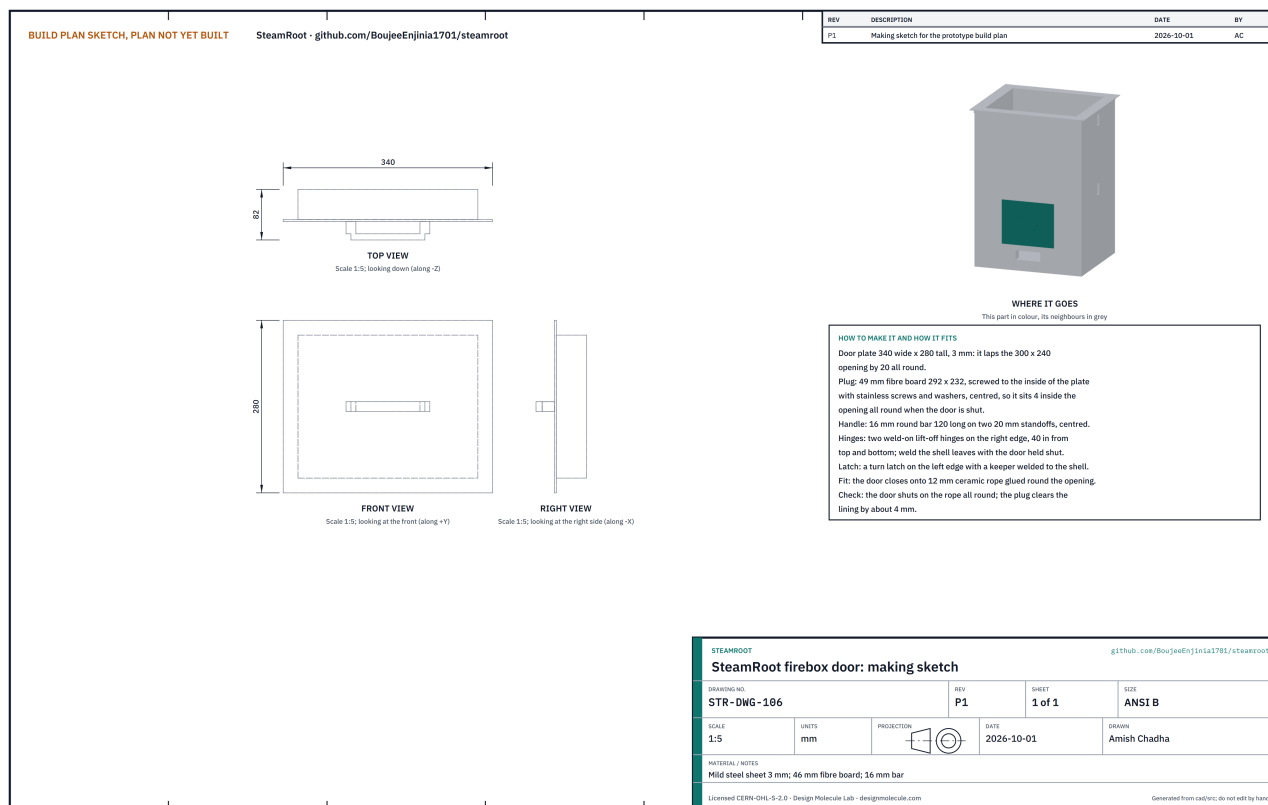


Figure 10. Firebox door making sketch (STR-DWG-106).

What it is and what it is made from. The door wood is loaded through, with a fibre board plug that fills the opening in the lining. Mild steel sheet 3 mm, 49 mm fibre board, 16 mm round bar, two weld-on lift-off hinges and a turn latch.

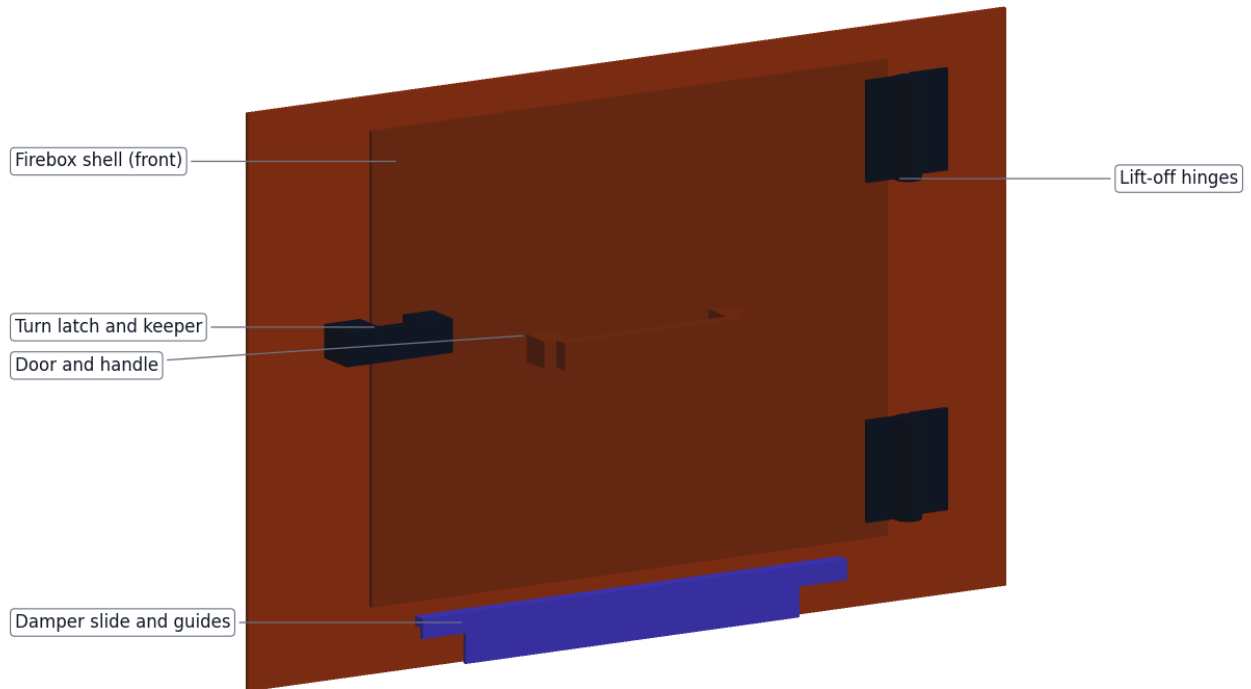
How to make it.

1. Cut the door plate 340 wide x 280 tall; it laps the opening by 20 all round.
2. Cut the plug from fibre board, 292 x 232 x 49, and screw it to the inside of the plate, centred, with stainless screws and large washers. It sits 4 inside the opening all round.
3. Weld a 120 handle of 16 mm bar on two 20 mm standoffs, centred on the front.
4. Weld the door leaves of the two hinges to the right edge of the door, 40 in from top and bottom. Hold the door shut in place and weld the shell leaves to the shell.
5. Fit the turn latch on the left edge of the door and weld its keeper to the shell.
6. Glue 12 mm ceramic rope round the opening on the shell, where the door plate closes.

How it fits the parts next to it.

Joint 9: door hinges, latch and air damper

Seen from the front left. The door hinges on its right edge and latches on its left



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Figure 11. The door hinges on its right edge and latches on its left. Below it, the air damper slides sideways between its guides.

Check before moving on. The door shuts on the rope all round and latches; the plug clears the lining by about 4.

3.8 Air damper slide and guides

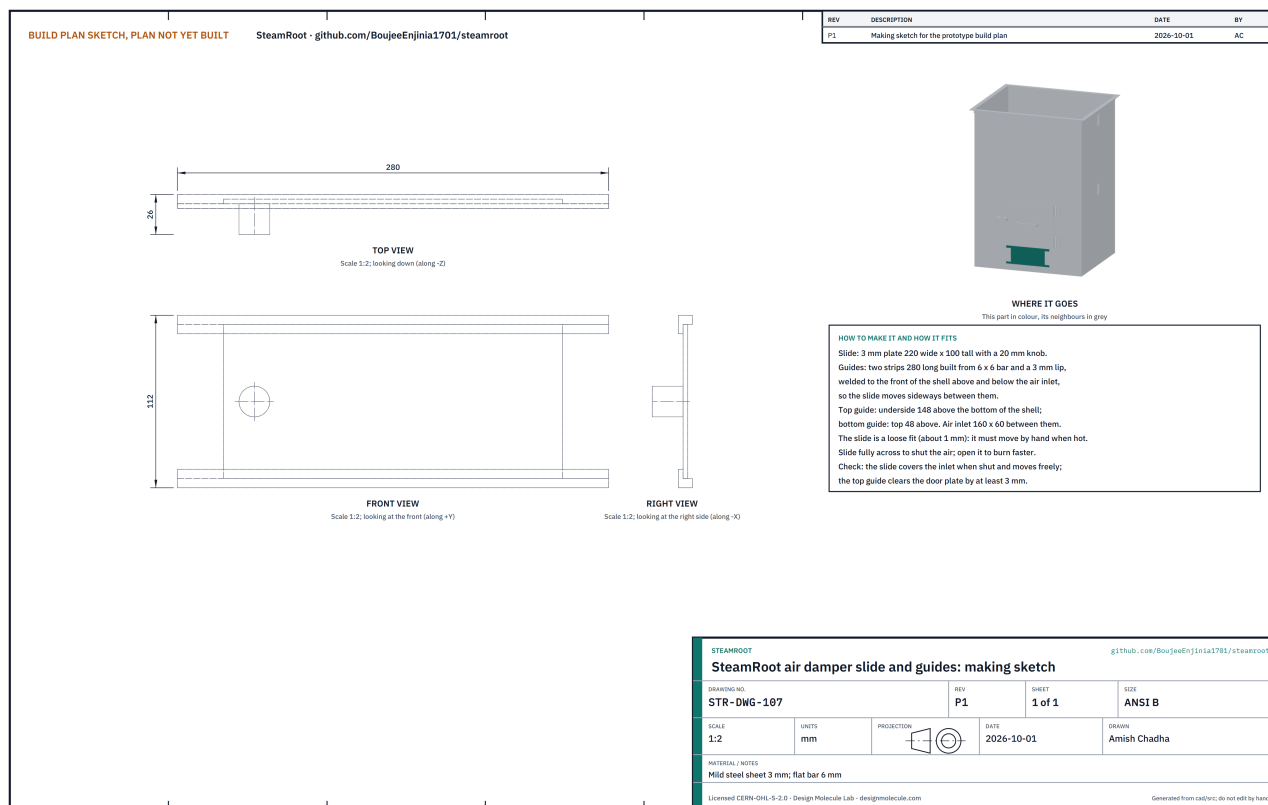


Figure 12. Air damper slide and guides making sketch (STR-DWG-107).

What it is and what it is made from. A plate that slides sideways over the air inlet to set how fast the fire burns. Mild steel sheet 3 mm, 6 mm bar.

How to make it.

1. Cut the slide 220 wide x 100 tall from 3 mm sheet and weld a 20 mm knob near its left end.
2. Make two guides 280 long, each from a 6 x 6 bar welded to the shell and a 3 mm lip over it, so they form a channel the slide runs in.
3. Weld the guides to the front of the shell, centred on the air inlet: the top guide's underside 148 above the bottom of the shell, the bottom guide's top 48 above. Check the top guide clears the door plate by at least 3.

How it fits the parts next to it. The slide is a loose fit, about 1 mm, so it moves by hand when hot (Figure 11). Fully across, it shuts the air off.

Check before moving on. The slide covers the inlet when shut and moves freely.

3.9 Monotube coil

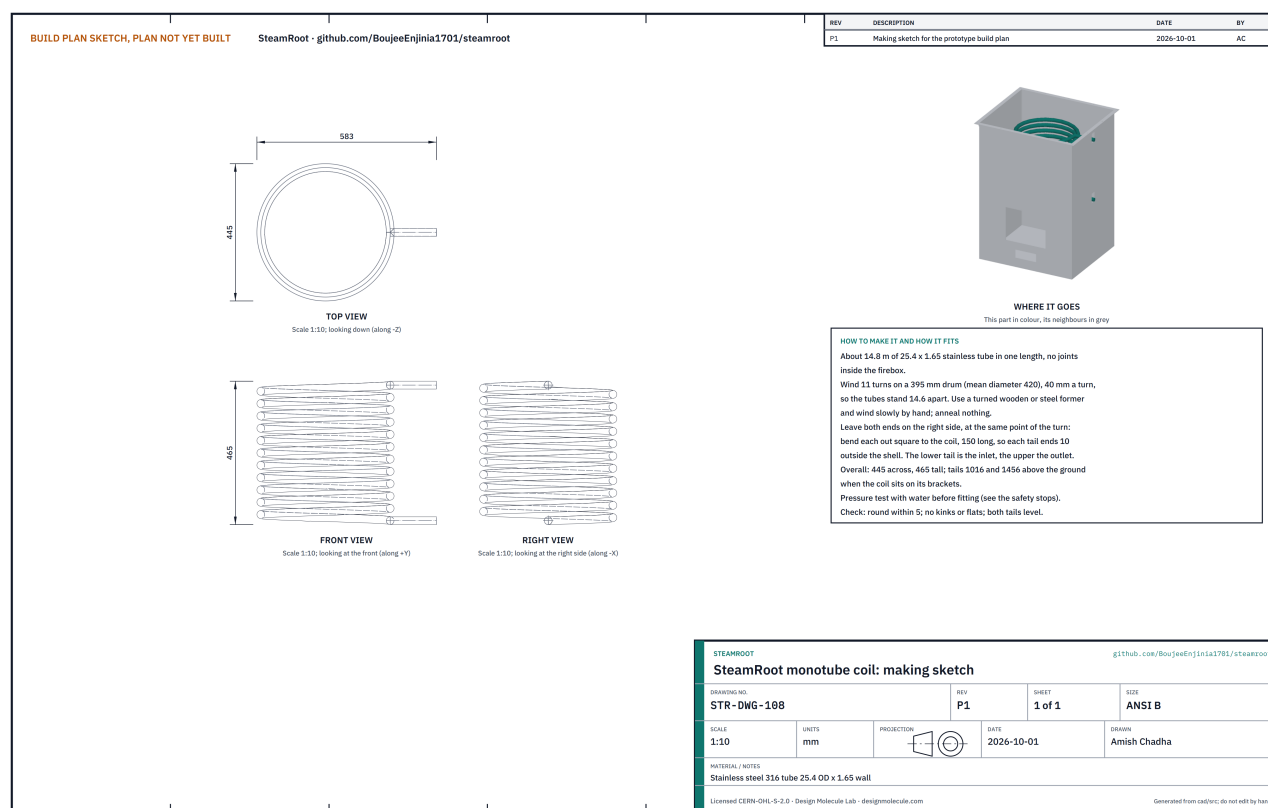


Figure 13. Monotube coil making sketch (STR-DWG-108).

What it is and what it is made from. The coil the water boils in, wound from one length of stainless steel 316 tube, 25.4 outside diameter, 1.65 wall, with no joint inside the firebox.

How to make it.

1. Make a former: a drum 395 in diameter and about 500 long, turned from timber or rolled from steel, with a slot to start the tube.
2. Leave 150 of straight tube at the start. Wind 11 turns by hand, slowly, keeping the turns 40 apart (14.6 between tubes) with spacer blocks. The mean diameter is 420.
3. Leave 150 of straight tube at the end. Bend each end out square to the coil on about a 75 radius, so both tails point the same way, one at the bottom and one at the top of the coil, and each ends 10 outside the shell when the coil is in place.
4. Slide the coil off the former. Check it is 445 across and 465 tall.
5. Fill it with water and pressure test it cold to 3 bar, the pump's shut-off pressure; hold 10 minutes with no fall and no leak.

How it fits the parts next to it. The coil rests on the three brackets (Figure 6). Its lower tail is the water inlet, 1016 above the ground; its upper tail is the steam outlet, 1456 above the ground. Both pass out through the slots in the right end (Figure 15).

Check before moving on. The coil is round within 5, with no kinks or flats, and both tails are level and parallel.

3.10 Tube gland plates (make 2)

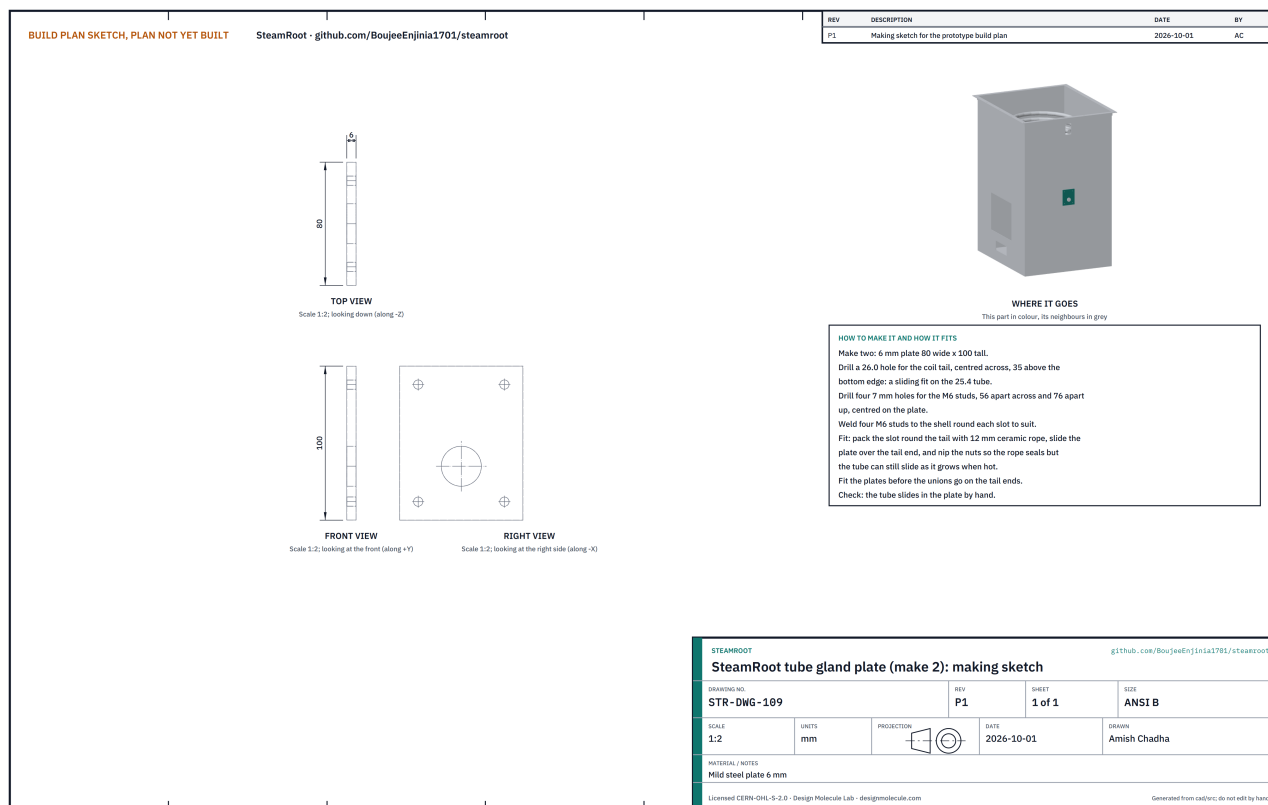


Figure 14. Tube gland plate making sketch (STR-DWG-109).

What it is and what it is made from. A plate that closes each tube slot round a coil tail and holds the rope packing that seals it, while letting the tube slide as it grows. Mild steel plate 6 mm.

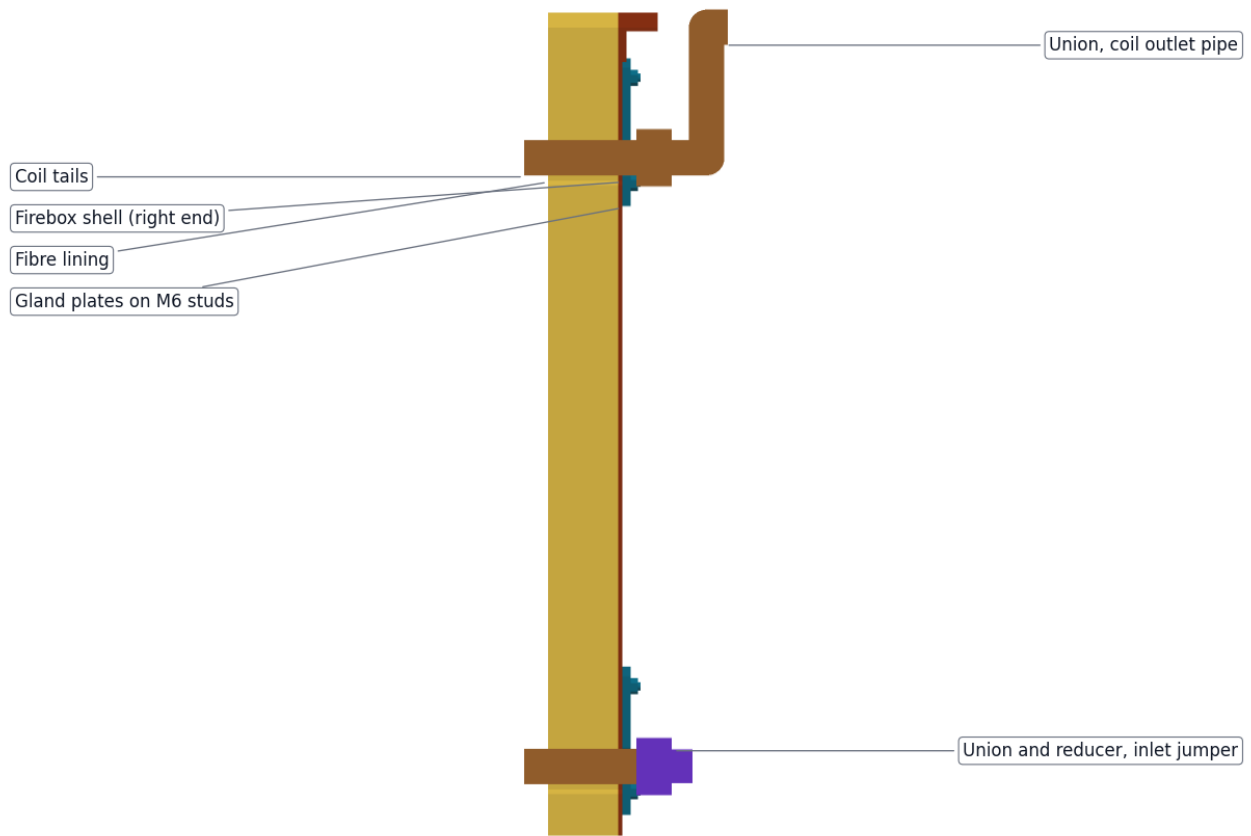
How to make it.

1. Cut two plates 80 wide x 100 tall.
2. Drill a 26.0 hole for the tail, centred across, 35 above the bottom edge.
3. Drill four 7 mm holes for the studs, 56 apart across and 76 apart up and down, centred on the plate. Mark the four M6 studs on the shell through these holes and weld them on (section 3.2).

How it fits the parts next to it.

Joint 4: coil tails out through the wall slots

Cut through the tails. Each slot runs 30 mm above its tail; the gland plate covers it and rope packing seals it



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Figure 15. Each slot runs 30 above its tail so the coil can go in high. Pack the slot round the tail with 12 mm ceramic rope, slide the plate over the tail end and nip the nuts. The unions go on afterwards.

Check before moving on. Each tail slides in its plate by hand.

3.11 Firebox roof and roof board

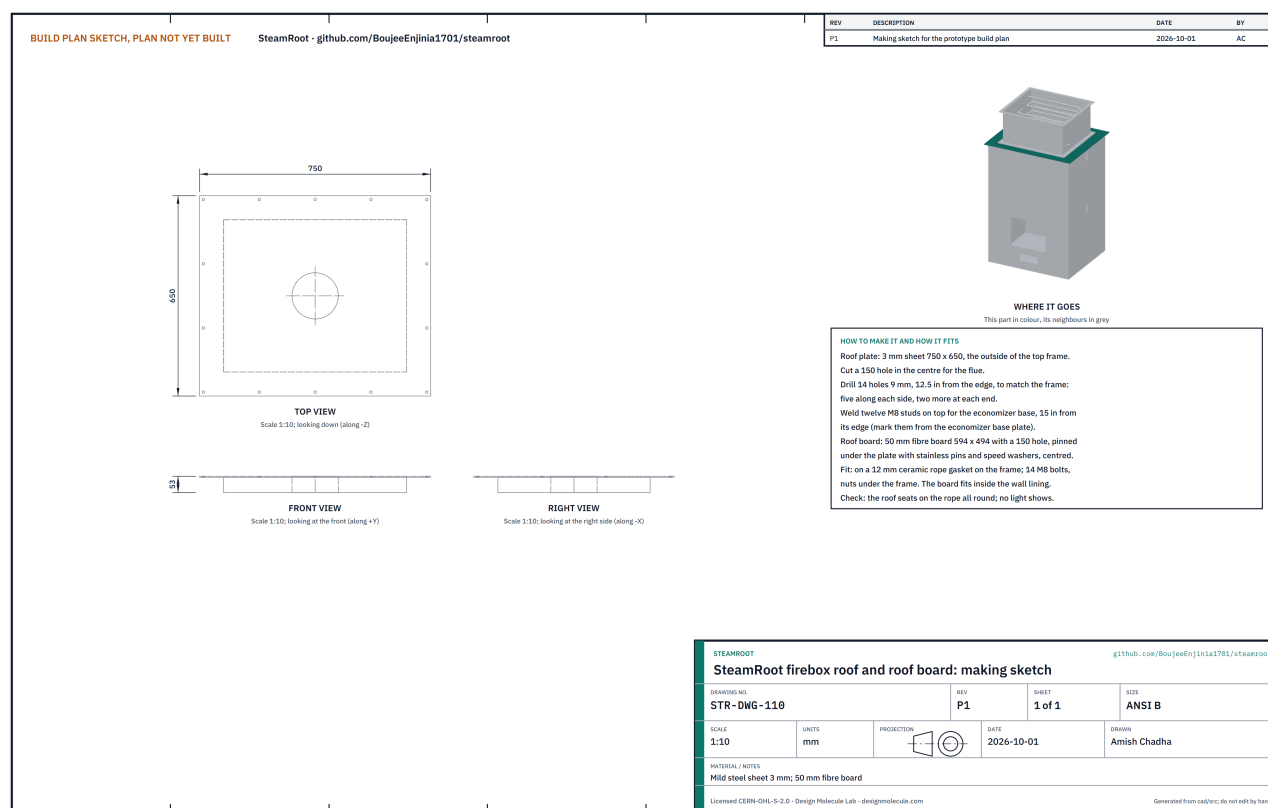


Figure 16. Firebox roof making sketch (STR-DWG-110).

What it is and what it is made from. The bolted lid of the firebox, lined underneath, which carries the economizer. Mild steel sheet 3 mm and 50 mm fibre board.

How to make it.

1. Cut the plate 750 x 650 and a 150 hole in its centre.
2. Drill 14 holes of 9 mm to match the top frame (clamp the plate to the frame and drill through).
3. Weld twelve M8 studs on top for the economizer base (mark them through the economizer base plate, section 3.12).
4. Pin the roof board (section 3.5) centred under the plate with stainless pins and speed washers, lining up the two 150 holes.

How it fits the parts next to it. It sits on a 12 mm ceramic rope gasket on the top frame, held by 14 M8 bolts with nuts under the frame; the board fits inside the tops of the wall lining (Figure 18).

Check before moving on. The roof seats on the rope all round and no light shows between them.

3.12 Economizer box

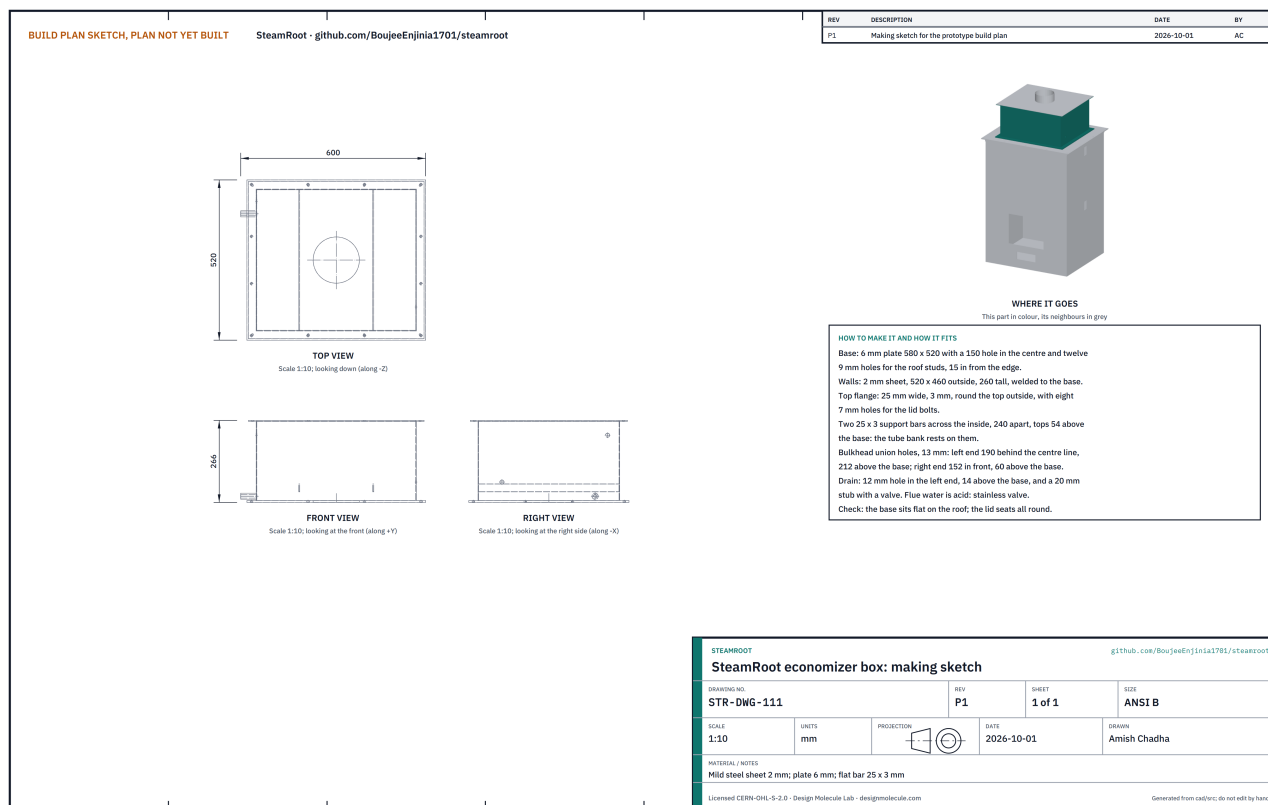


Figure 17. Economizer box making sketch (STR-DWG-111).

What it is and what it is made from. The box above the firebox that the hot flue gas rises through, around the tube bank that warms the feed water. Mild steel sheet 2 mm, 6 mm plate and 25 x 3 mm flat bar.

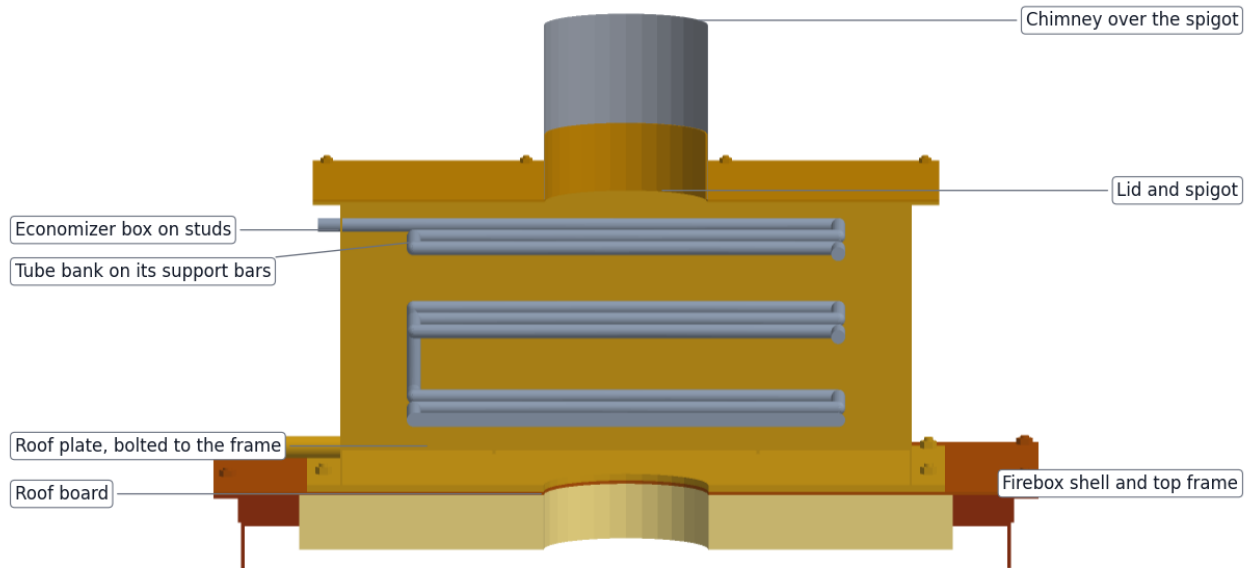
How to make it.

1. Cut the base 580 x 520 from 6 mm plate, with a 150 hole in the centre and twelve 9 mm holes 15 in from the edge for the roof studs.
2. Cut and weld the walls from 2 mm sheet, 520 x 460 outside and 260 tall, onto the base. Weld a 25 wide, 3 mm flange round the top outside, with eight 7 mm holes for the lid bolts.
3. Weld two 25 x 3 support bars across the inside, 240 apart, their tops 54 above the base.
4. Drill the two 13 mm holes for the bulkhead unions: in the left end 190 behind the centre line and 212 above the base; in the right end 152 in front and 60 above the base. Fit the unions.
5. Drill a 12 mm drain hole in the left end, 14 above the base, 150 behind the centre line, and fit a stainless drain valve. The condensate is acid.

How it fits the parts next to it.

Joint 5: roof, economizer and chimney, cut through the middle

Seen from the front. Flue gas rises through the roof hole, past the tube bank and up the chimney



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Figure 18. The base sits on a rope gasket on the roof over the twelve studs, nuts on top. Flue gas rises through the roof hole, past the tube bank and up the chimney.

Check before moving on. The base sits flat on the roof; the support bars are level.

3.13 Economizer tube bank

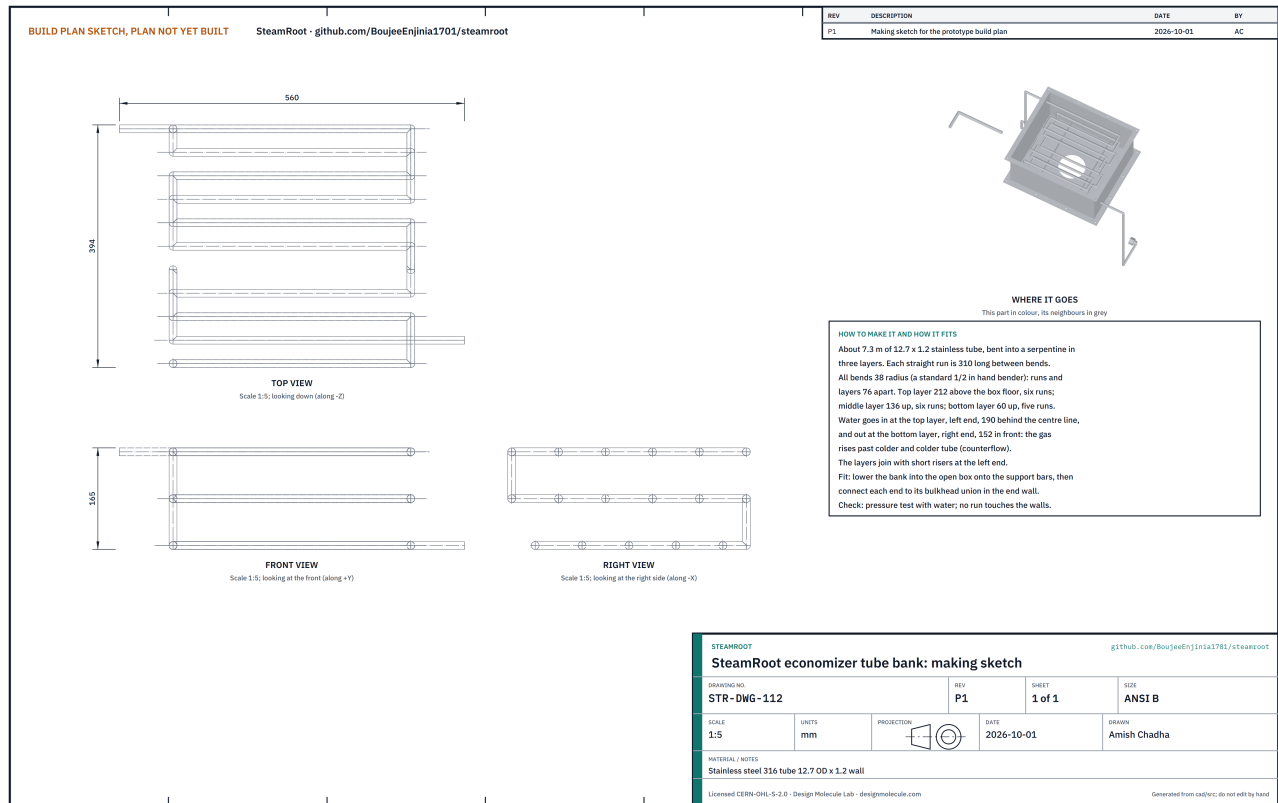


Figure 19. Economizer tube bank making sketch (STR-DWG-112).

What it is and what it is made from. The serpentine of stainless tube inside the economizer that the feed water flows through. About 7.3 m of stainless steel 316 tube, 12.7 outside diameter, 1.2 wall.

How to make it.

1. Bend the tube with a standard 1/2 in hand bender (38 radius), so the straight runs are 76 apart and each is 310 long between bends.
2. Top layer, 212 above the box floor: six runs, starting at the left end 190 behind the centre line. Turn down at the left end to the middle layer, 136 above the floor: six runs. Turn down again to the bottom layer, 60 above the floor: five runs, ending at the right end 152 in front of the centre line.
3. Leave a short straight at each end to meet the bulkhead unions.
4. Pressure test it cold to 3 bar for 10 minutes.

How it fits the parts next to it. Lower it into the open box onto the two support bars, then connect each end to its bulkhead union (step 10). Water enters at the top and leaves at the bottom, so the rising gas meets colder and colder tube.

Check before moving on. No run touches a wall; no leak at 3 bar.

3.14 Economizer lid

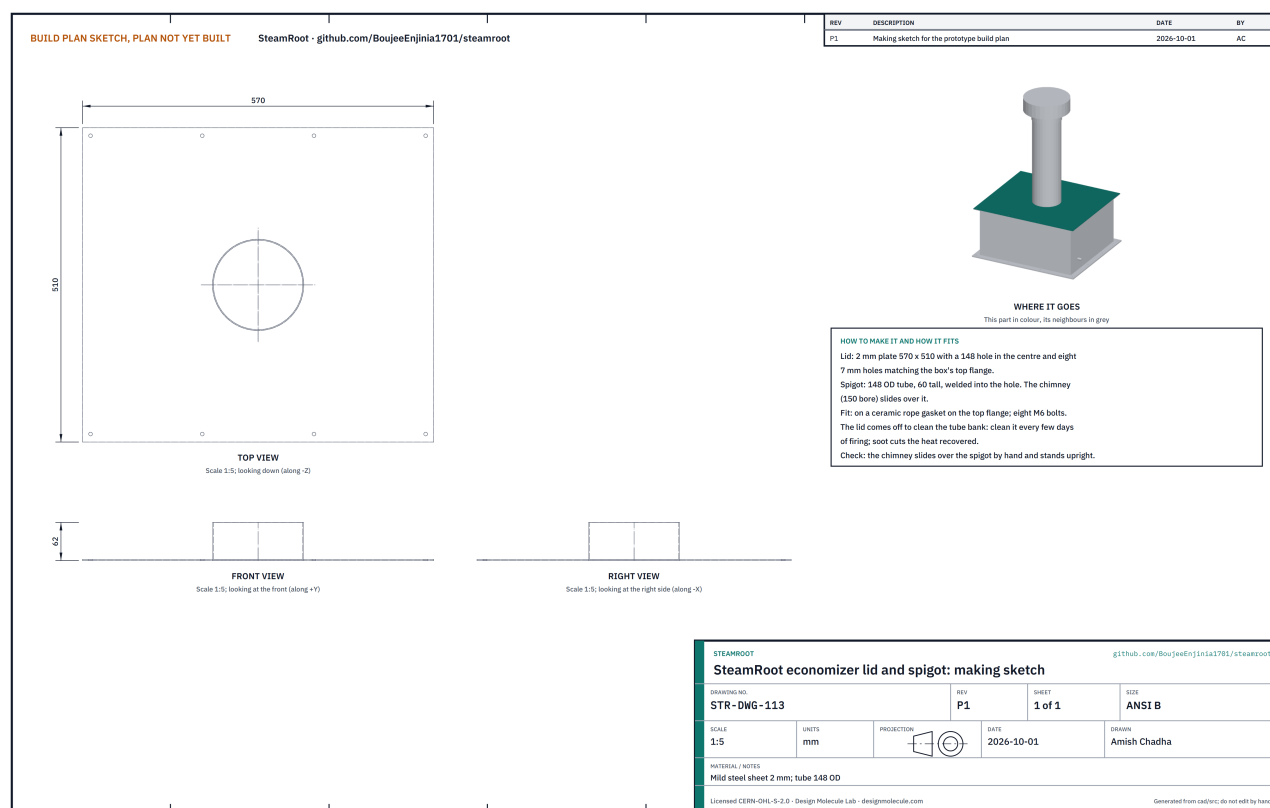


Figure 20. Economizer lid and spigot making sketch (STR-DWG-113).

What it is and what it is made from. The bolted top of the economizer, carrying the spigot the chimney stands on. Mild steel sheet 2 mm and a short length of 148 mm tube.

How to make it.

1. Cut the lid 570 x 510 with a 148 hole in the centre and eight 7 mm holes matching the box's top flange.
2. Weld a 148 outside diameter tube, 60 tall, upright into the hole.

How it fits the parts next to it. The lid sits on a rope gasket on the top flange under eight M6 bolts; the chimney (150 bore) slides over the spigot (Figure 18). Take the lid off to brush soot from the tube bank.

Check before moving on. The chimney slides over the spigot by hand and stands upright.

3.15 Header post

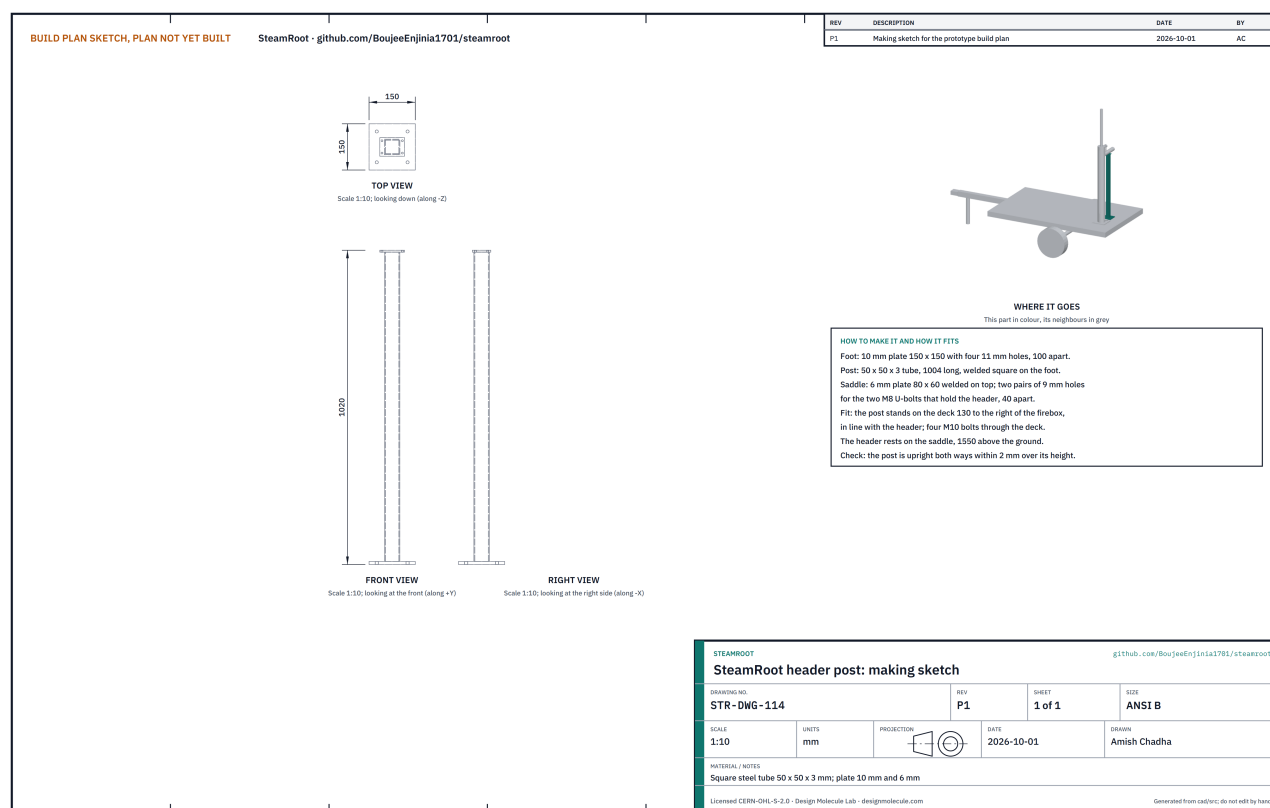


Figure 21. Header post making sketch (STR-DWG-114).

What it is and what it is made from. The post that carries the steam header 1550 above the ground. Square steel tube 50 x 50 x 3 mm, 10 mm and 6 mm plate, two M8 U-bolts.

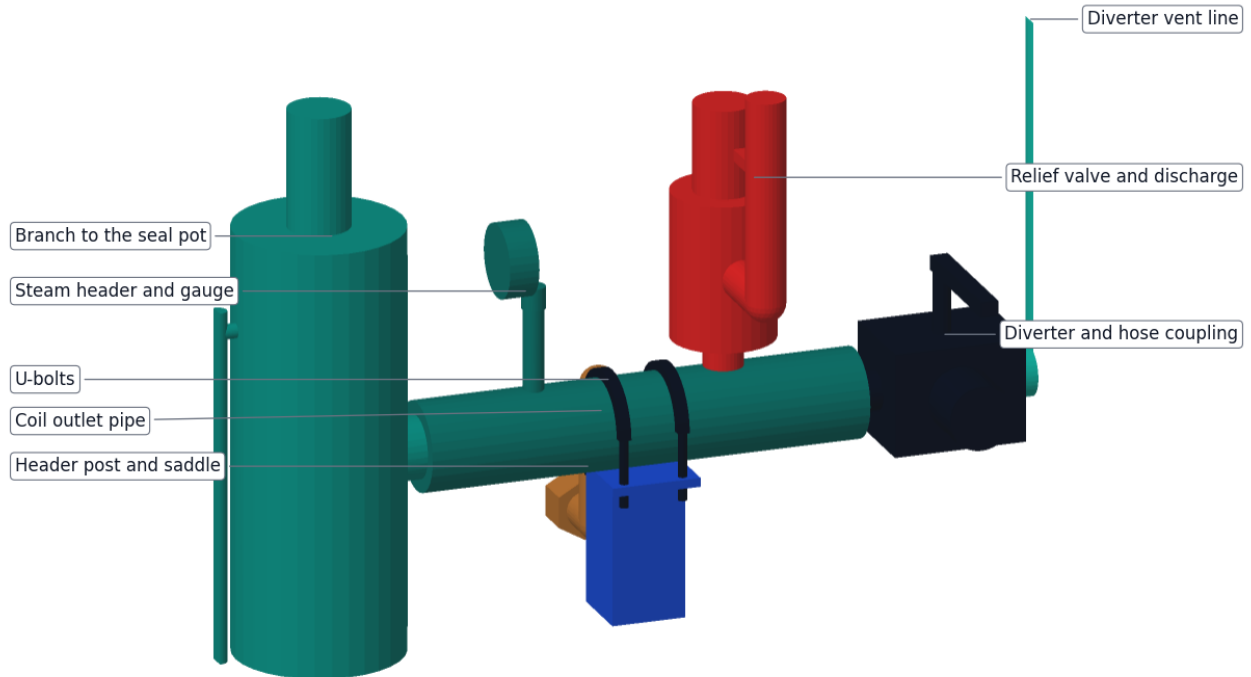
How to make it.

1. Cut the foot 150 x 150 from 10 mm plate and drill four 11 mm holes 100 apart.
2. Cut the post 1004 long and weld it square on the centre of the foot.
3. Weld a 6 mm saddle plate 80 x 60 on top and drill two pairs of 9 mm holes for the U-bolts, 40 apart.

How it fits the parts next to it.

Joint 6: the steam header on its post

Seen from the right and above. Coil steam comes in from the left; the seal pot branch can never be closed



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Figure 22. The header lies on the saddle under two U-bolts. The coil outlet pipe enters its left side, the seal pot branch leaves its front end, the diverter its back end; the relief valve and gauge stand on top.

The post stands on the deck 130 to the right of the firebox, on the centre line, held by four M10 bolts through the deck.

Check before moving on. The post is upright both ways within 2 over its height.

3.16 Steam header

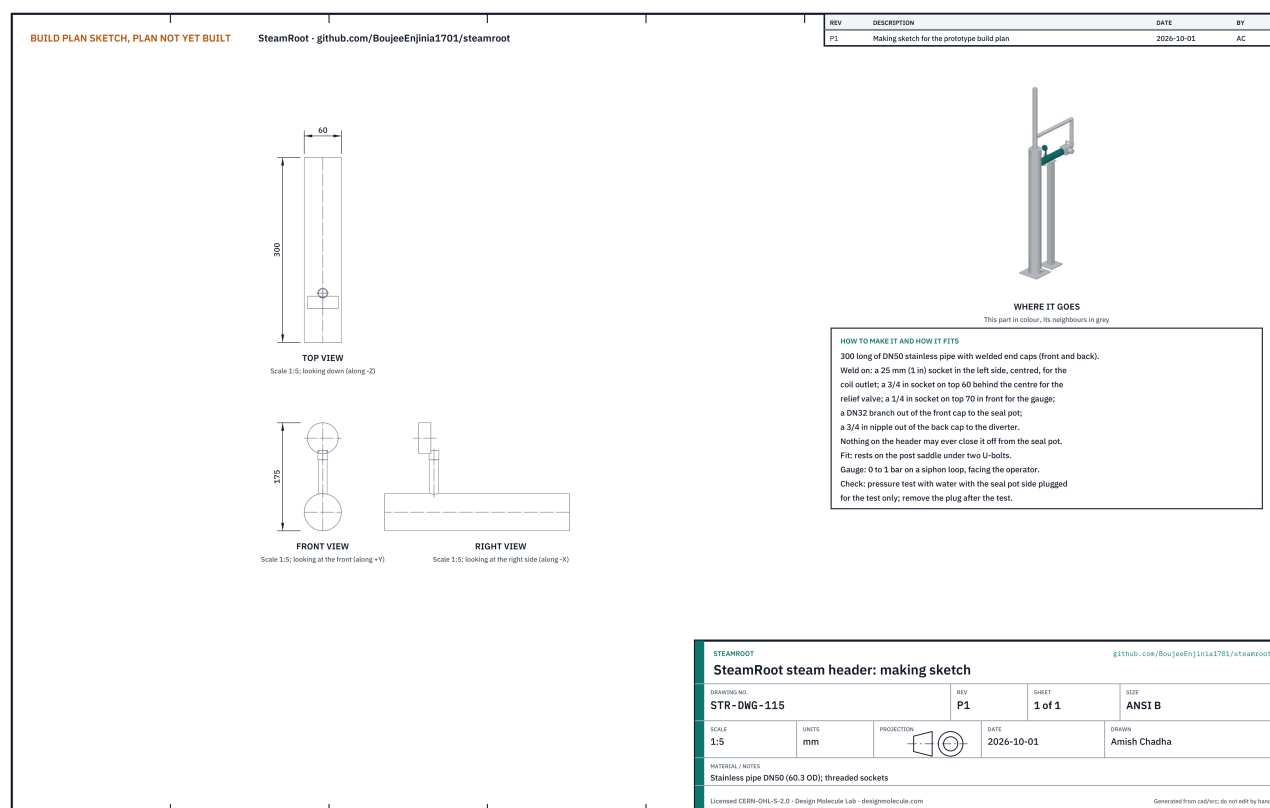


Figure 23. Steam header making sketch (STR-DWG-115).

What it is and what it is made from. A short, wide pipe that collects the steam from the coil and passes it to the seal pot, the relief valve, the gauge and the diverter. Stainless steel pipe DN50 (60.3 outside diameter), 300 long, with welded caps and sockets.

How to make it.

1. Cut 300 of DN50 pipe and weld a cap on each end.
2. Weld on: a 1 in socket in the left side, centred, for the coil outlet; a 3/4 in socket on top 60 behind the centre for the relief valve; a 1/4 in socket on top 70 in front of the centre for the gauge; a DN32 branch out of the front cap toward the seal pot; a 3/4 in nipple out of the back cap for the diverter.
3. Fit the gauge (0 to 1 bar) on a siphon loop, facing the front.

How it fits the parts next to it. See Figure 22. Nothing fitted to the header may ever close it off from the seal pot.

Check before moving on. Pressure test it cold to 3 bar with the seal pot branch plugged for the test only; remove the plug at once afterwards.

3.17 Water-seal pot, dip leg and vent

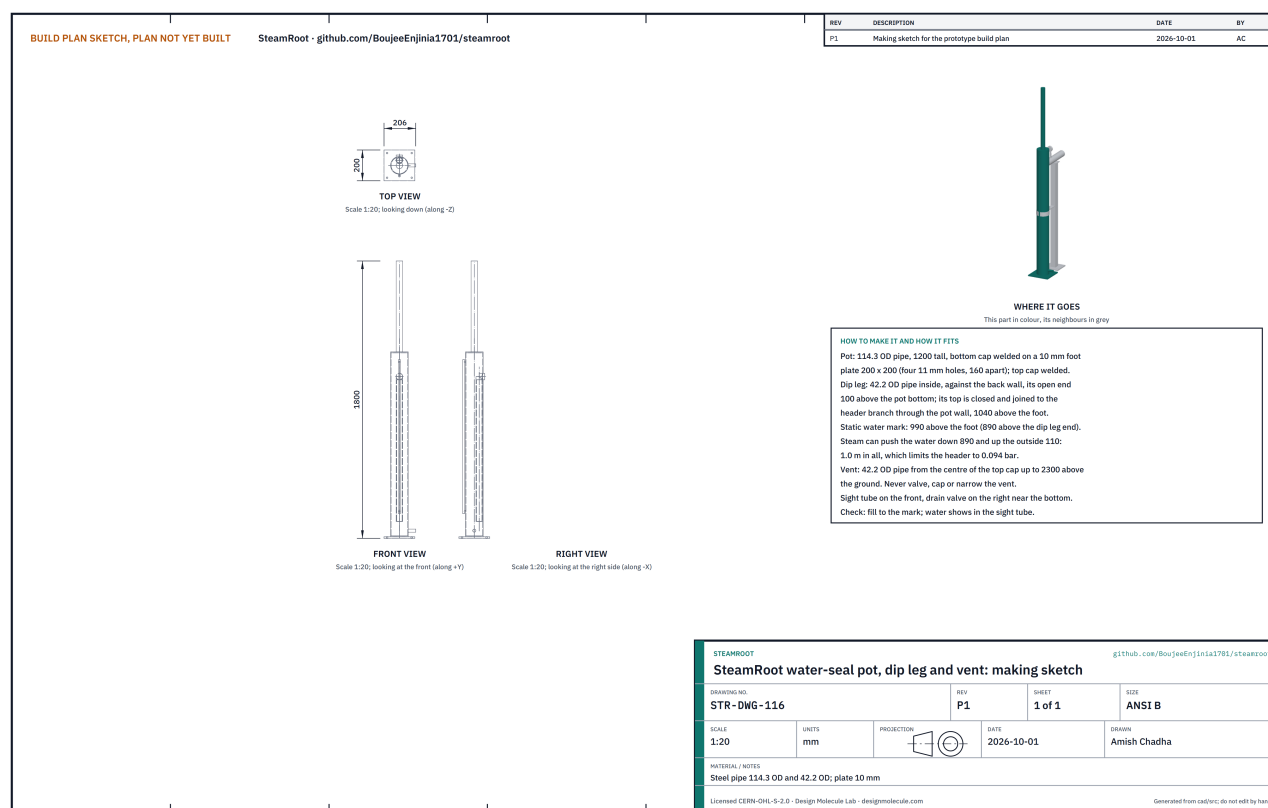


Figure 24. Water-seal pot, dip leg and vent making sketch (STR-DWG-116).

What it is and what it is made from. The pot of water that limits the steam pressure: steam from the header goes down a dip leg under the water, and if the pressure ever passes 0.094 bar it pushes the water out of the way and escapes up the vent. Steel pipe 114.3 and 42.2 outside diameter, 10 mm plate.

How to make it.

1. Cut the pot 1200 long from 114.3 pipe; weld a cap on the bottom and stand it on a 10 mm foot plate 200 x 200 with four 11 mm holes 160 apart.
2. Cut the dip leg from 42.2 pipe, open at the bottom and closed at the top. Fit it inside the pot against the back wall, its open end 100 above the pot bottom, and join its top through the pot wall to the header branch, 1040 above the foot plate.
3. Weld the top cap on, and the vent pipe (42.2) up from its centre to 2300 above the ground.
4. Fit a level sight tube on the front, with the static water mark 990 above the foot plate (890 above the dip leg's open end), and a drain valve on the right near the bottom.

How it fits the parts next to it.

Joint 7: the water-seal pot, cut open

Steam from the header enters the dip leg. Over 0.094 bar it pushes the water down 890 mm and bubbles out to the vent

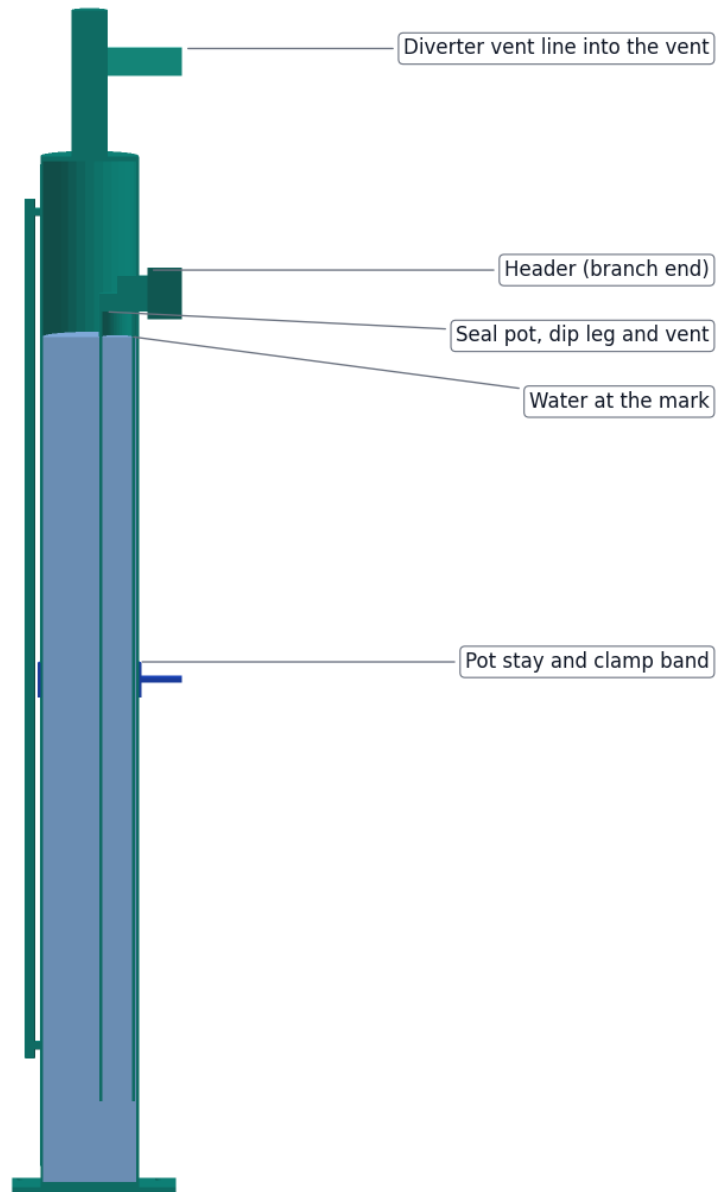


Figure 25. With water at the mark, steam can push the water 890 down the dip leg, which raises the water outside it by 110: 1.0 m of water in all, 0.094 bar. Above that, steam bubbles through and leaves by the vent. The diverter's vent line joins the vent above the pot.

The pot stands on the deck in front of the header post, held by four M10 bolts and by the stay (section 3.18).

Check before moving on. Filled to the mark, the water shows in the sight tube; the vent is open from end to end.

3.18 Pot stay

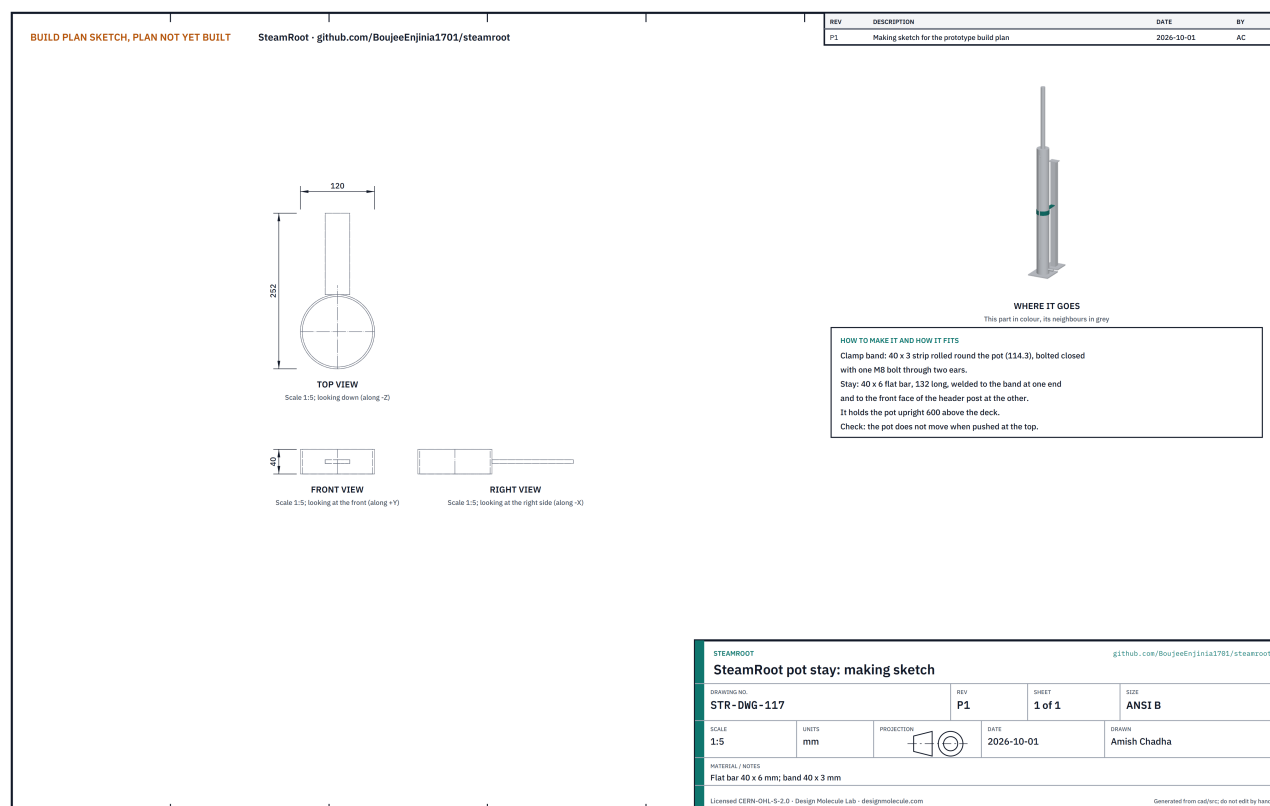


Figure 26. Pot stay making sketch (STR-DWG-117).

What it is and what it is made from. A short bar that ties the tall seal pot to the header post. Flat bar 40 x 6 mm and a 40 x 3 mm band.

How to make it.

1. Roll a 40 x 3 band round a piece of 114.3 pipe, with two ears for an M8 bolt.
2. Cut the stay 132 long from 40 x 6 bar and weld it to the band.

How it fits the parts next to it. Bolt the band round the pot 600 above the deck and weld the other end of the stay to the front of the post (Figure 22).

Check before moving on. The pot does not move when pushed at the top.

3.19 Drum saddles (make 2)

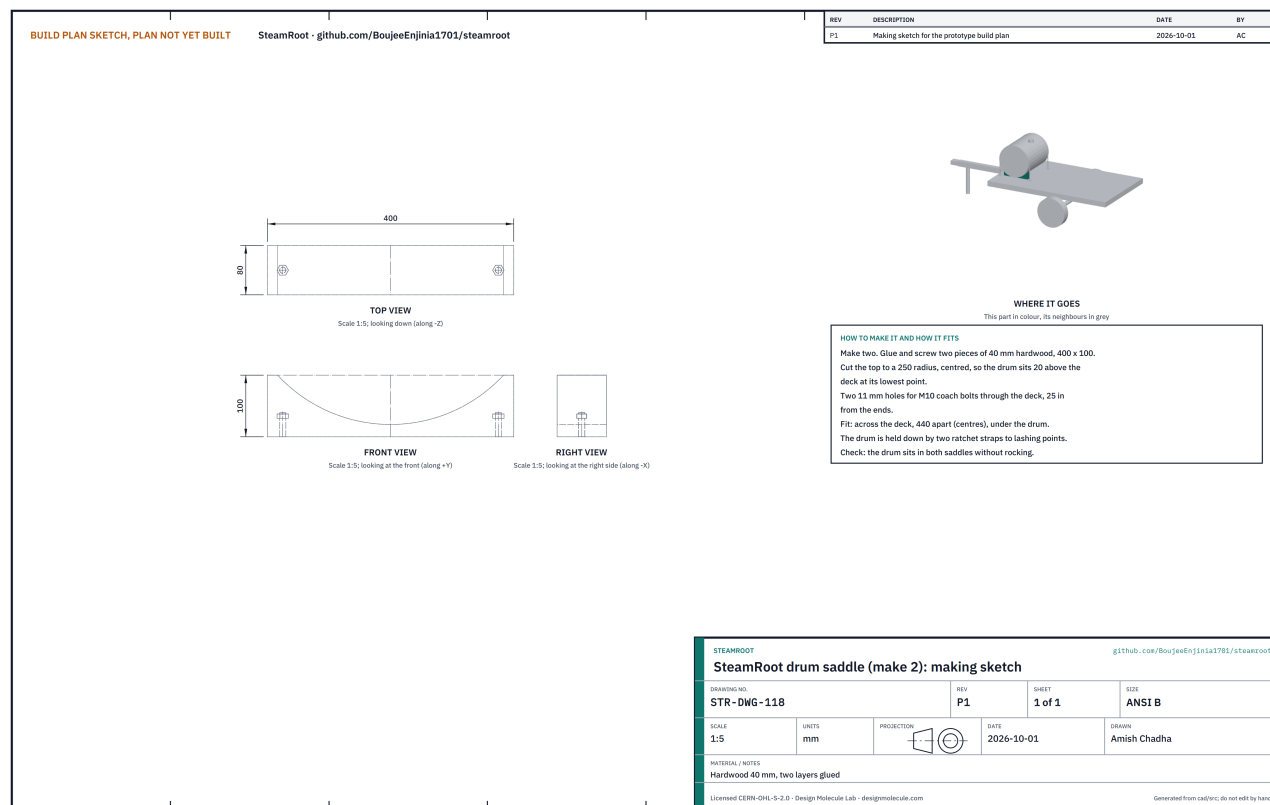


Figure 27. Drum saddle making sketch (STR-DWG-118).

What it is and what it is made from. Two curved blocks the feed drum lies in. Hardwood 40 mm, two layers glued and screwed.

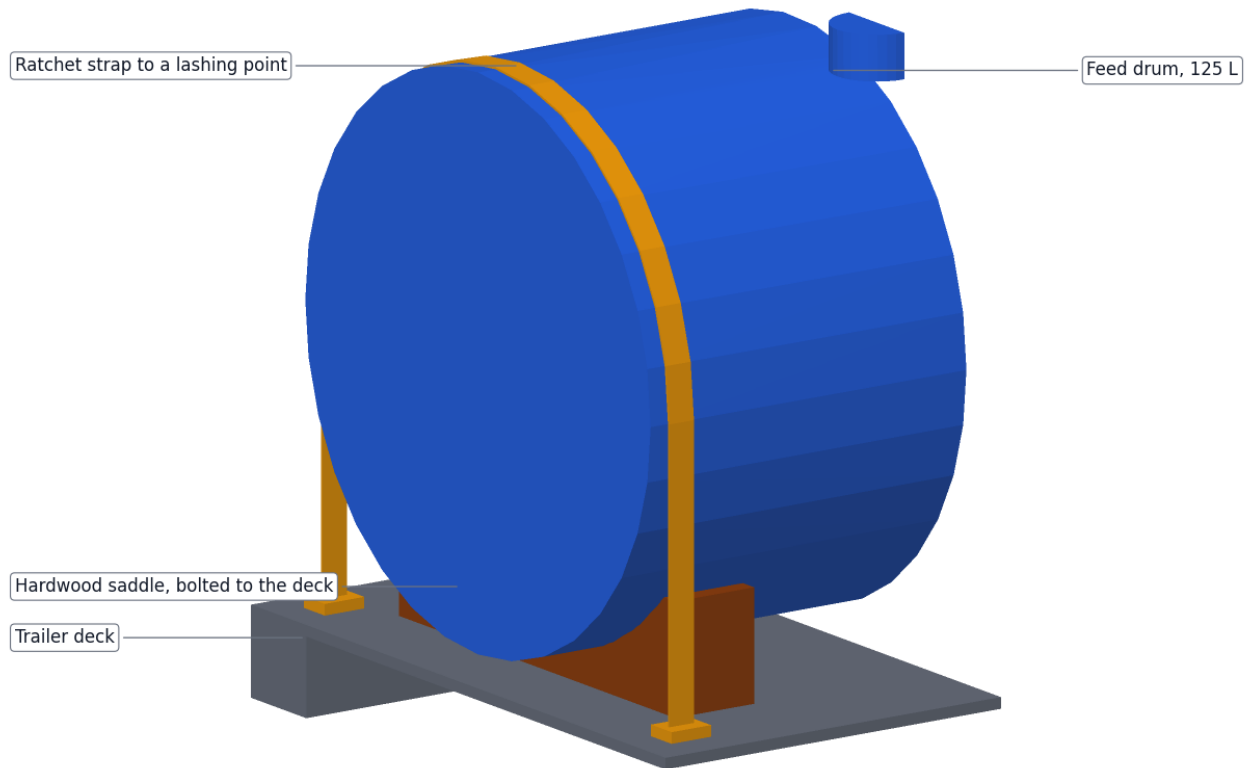
How to make it.

1. Glue and screw two pieces of 40 mm hardwood, 400 x 100, for each saddle.
2. Cut the top to a 250 radius, centred, so the drum sits 20 above the deck at its lowest point.
3. Drill two 11 mm holes 25 in from the ends for M10 coach bolts.

How it fits the parts next to it.

Joint 10: feed drum on its saddles (front half)

The drum lies across the deck in two curved saddles; a ratchet strap over each holds it down



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Figure 28. The drum lies across the deck in the two saddles, 440 apart, held down by a ratchet strap over each to lashing points on the deck.

Check before moving on. The drum sits in both saddles without rocking.

3.20 Hood shells (make 2)

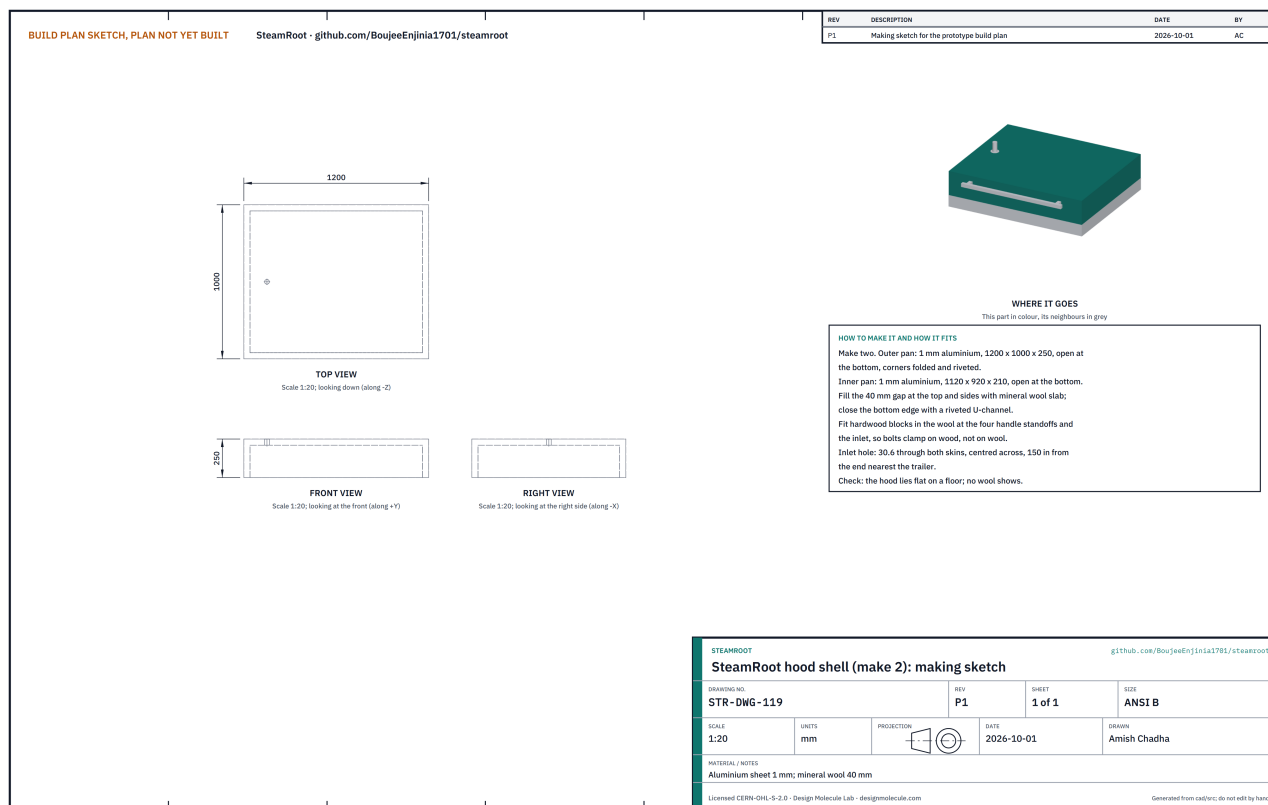


Figure 29. Hood shell making sketch (STR-DWG-119).

What it is and what it is made from. The insulated open-bottomed pan that is pressed onto the soil. Aluminium sheet 1 mm, two skins, with 40 mm mineral wool slab between them.

How to make it.

1. Fold the outer pan from 1 mm aluminium, 1200 x 1000 x 250, open at the bottom, corners folded and riveted.
2. Fold the inner pan, 1120 x 920 x 210, the same way.
3. Glue hardwood blocks to the inside of the outer pan where the four handle standoffs and the inlet will go.
4. Lay 40 mm mineral wool slab in the outer pan, set the inner pan in, and close the bottom edge all round with a riveted aluminium U-channel.
5. Cut the inlet hole, 30.6 through both skins and the block, centred across, 150 in from the end nearest the trailer.

How it fits the parts next to it. The skirt, manifold and handles fit on it (Figure 32).

Check before moving on. The hood lies flat on a floor; no wool shows.

3.21 Hood skirts (make 2)

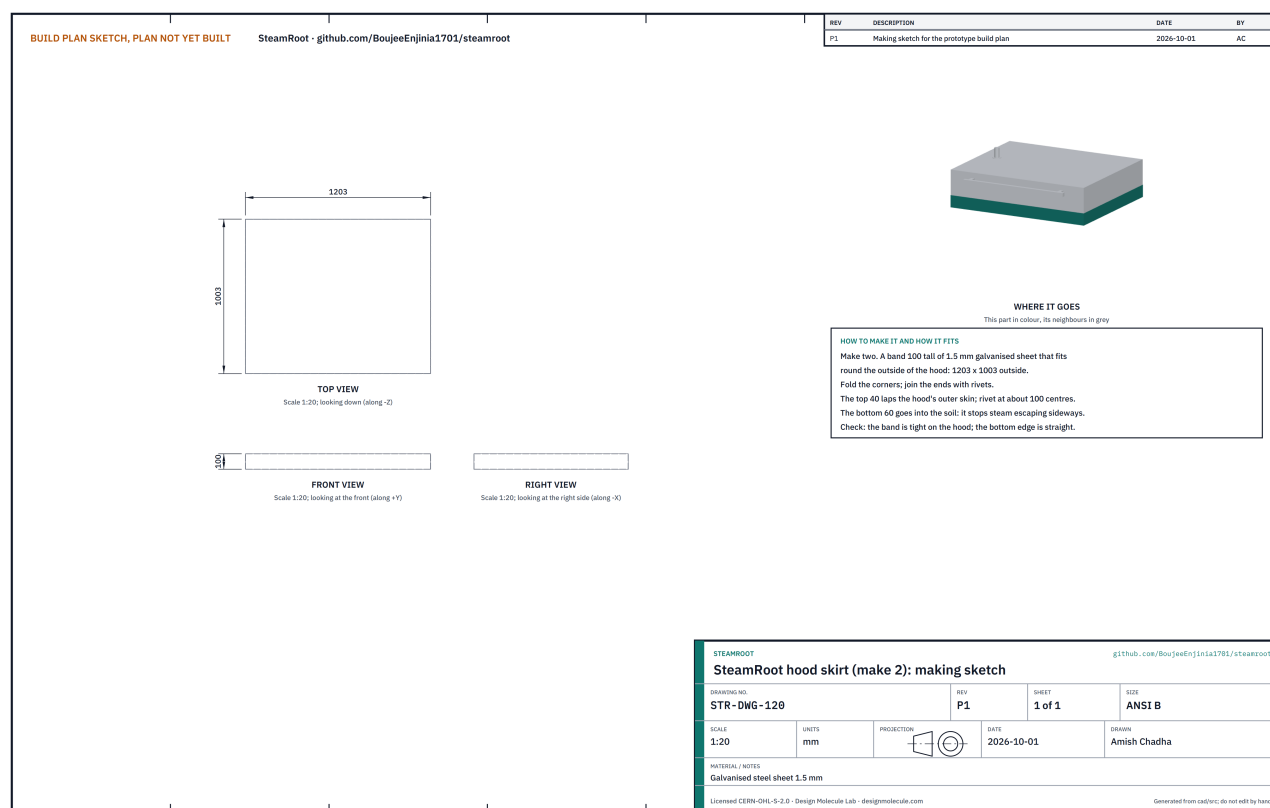


Figure 30. Hood skirt making sketch (STR-DWG-120).

What it is and what it is made from. A band round the bottom of the hood that goes 60 into the soil, so steam goes down, not sideways. Galvanised steel sheet 1.5 mm.

How to make it.

1. Cut a strip 100 wide and about 4420 long; fold the four corners so it fits round the hood, 1203 x 1003 outside.
2. Rivet the ends together.
3. Push it onto the hood so it laps the outer skin by 40, and rivet at about 100 centres. Do not weld galvanised steel.

How it fits the parts next to it. See Figure 32.

Check before moving on. The band is tight on the hood and its bottom edge is straight.

3.22 Hood manifolds (make 2)

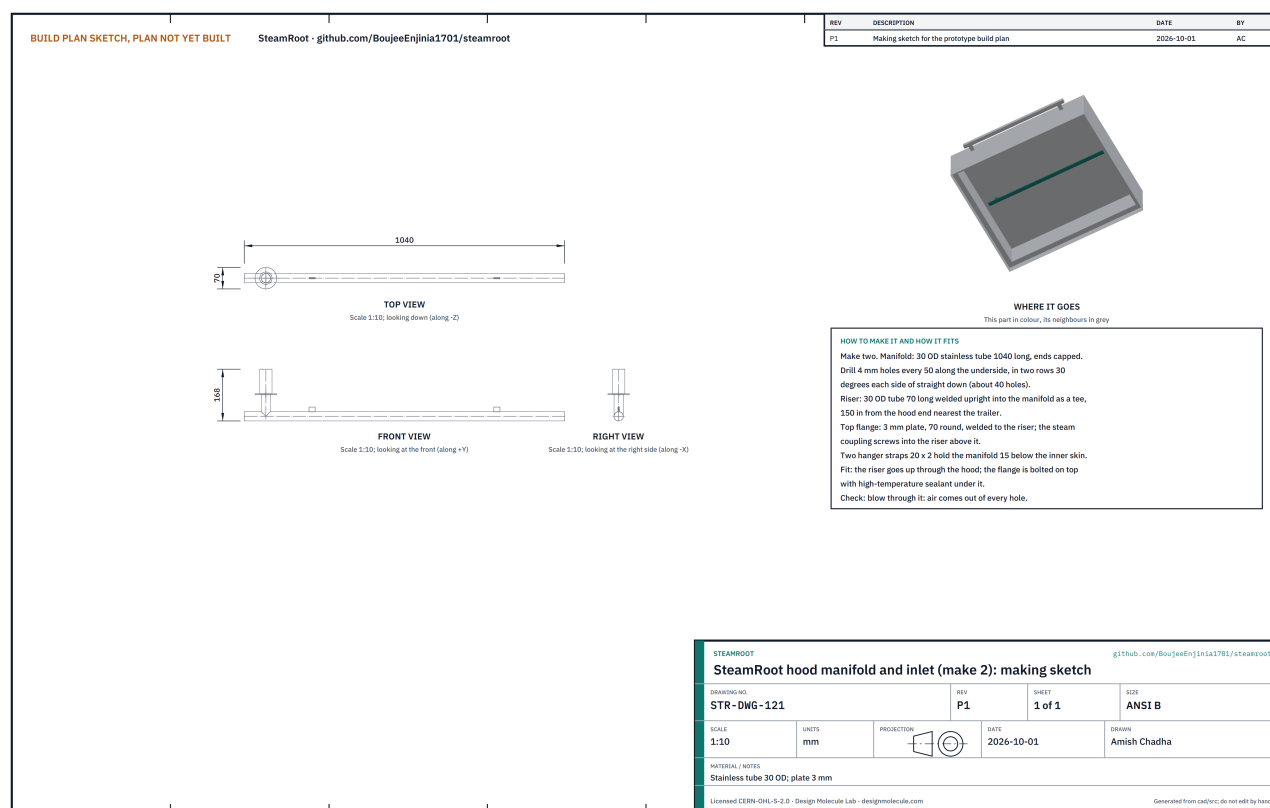


Figure 31. Hood manifold and inlet making sketch (STR-DWG-121).

What it is and what it is made from. The perforated pipe inside the hood that spreads the steam, with the riser and coupling the hose connects to. Stainless tube 30 outside diameter, 3 mm plate, a steam coupling.

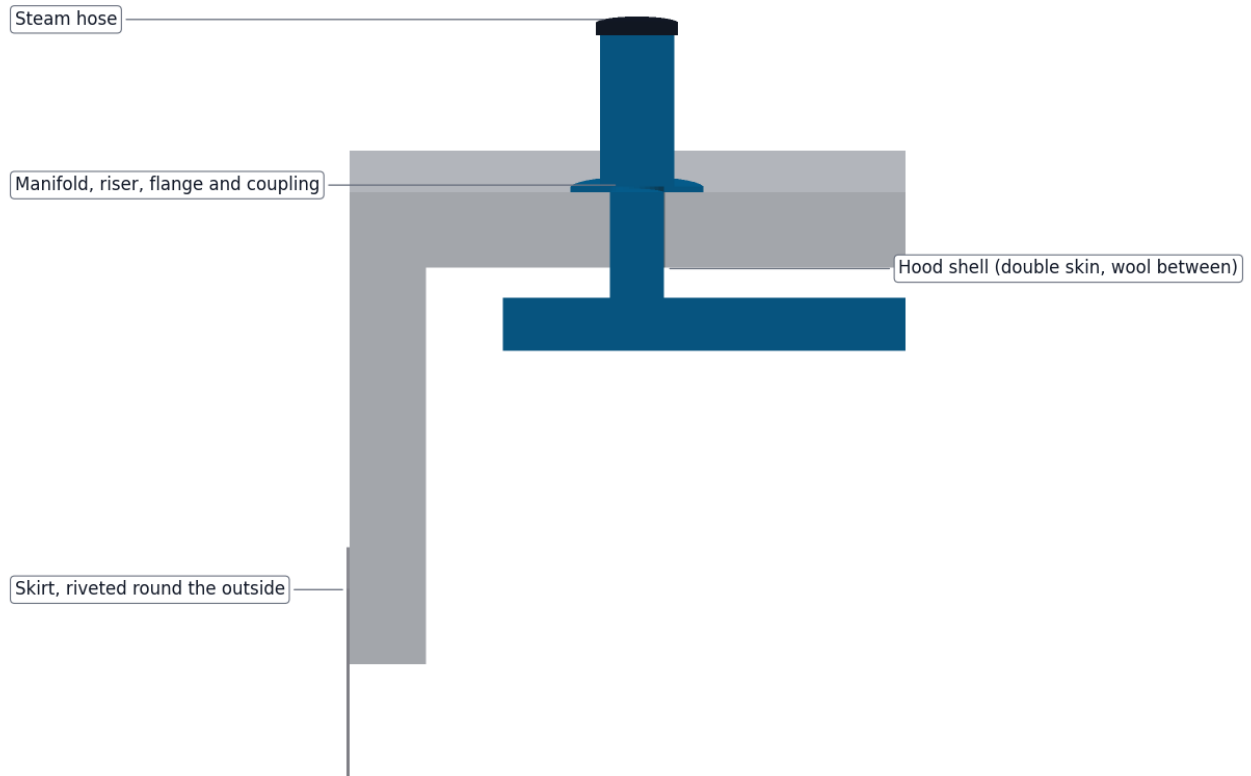
How to make it.

1. Cut the manifold 1040 long and cap both ends.
2. Drill 4 mm holes every 50 along the underside, in two rows 30 degrees either side of straight down.
3. Weld a 70 long riser upright into the manifold as a tee, 150 from the end nearest the trailer, and a 70 round, 3 mm flange on the riser 70 above the manifold's centre line.
4. Screw the steam coupling into the top of the riser.

How it fits the parts next to it.

Joint 8: hood end, cut through the steam inlet

Seen from the front. The skirt goes 60 mm into the soil; steam leaves the manifold through holes on its underside



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Figure 32. The riser passes up through the hood; the flange is bolted down on high-temperature sealant; two straps hang the manifold 15 below the inner skin. The skirt laps the outside of the hood.

Check before moving on. Blow through the coupling: air comes out of every hole.

3.23 Pump and alarm box and its wiring

SteamRoot prototype: 12 V wiring of the feed pump and alarms

Bought parts wired at block level inside the pump and alarm box. Stranded copper; ferrules or crimped lugs on every terminal.

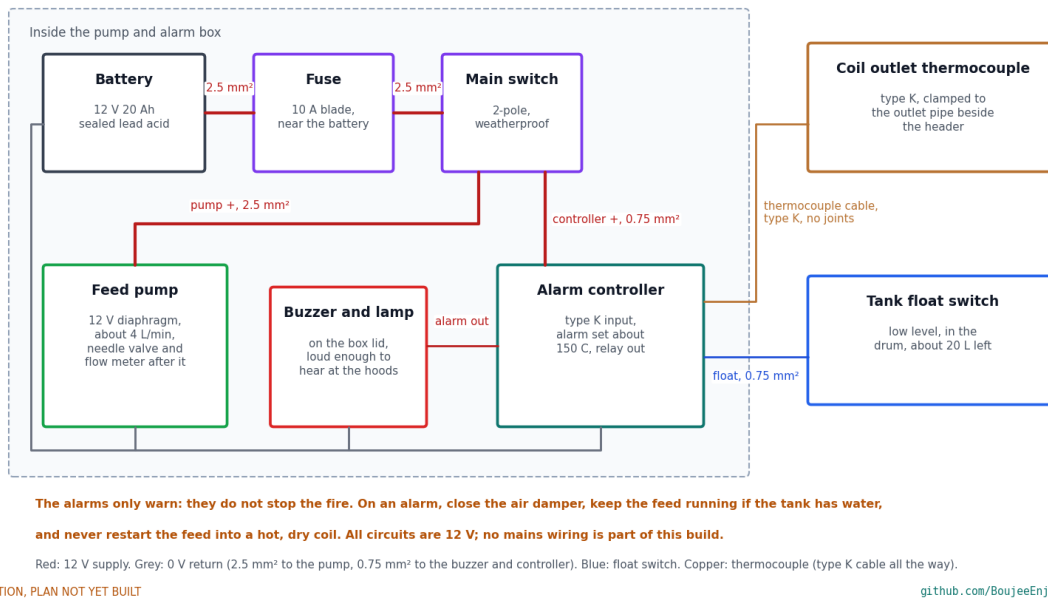


Figure 33. Block-level wiring of the 12 V parts. All circuits are 12 V from a sealed lead acid battery; there is no mains wiring.

What it is. A weatherproof box, about 240 x 250 x 240, bolted to the deck on the left of the firebox, holding the feed pump with its needle valve and flow meter, the 12 V battery, a fuse, the main switch and the alarm controller, with the buzzer and lamp on its lid.

What to do to it. Mount the parts on a board in the box and wire them as Figure 33 shows: battery through a 10 A fuse and the main switch to the pump and the controller in 2.5 mm² and 0.75 mm² wire; the type K thermocouple, clamped to the coil outlet pipe beside the header, back to the controller in type K cable with no joints; the drum's float switch to the controller; the controller's relay to the buzzer and lamp. Fit a cable gland for each cable that enters the box.

Check before moving on. With the main switch on, heating the thermocouple with a hot air gun sounds the alarm at about 150 °C, and lifting the float switch out of water sounds it too.

3.24 Bought components

Buy to specification, not brand. Line numbers are those of the bill of materials.

- **Trailer (line 1).** As section 3.1.
- **Feed drum (line 3).** 125 L HDPE drum with a bulkhead outlet and strainer near the bottom of one end, a filler cap and a level sight tube.
- **Feed pump (line 4).** 12 V diaphragm pump, about 4 L/min, shut-off pressure 3 bar or less, with a needle valve and a rotameter flow meter.
- **Lining, grate, door hardware (line 5).** Ceramic fibre blanket and board rated for 1260 °C with rigidizer; 300 x 300 cast grate; two weld-on lift-off hinges; a turn latch; 12 mm ceramic rope.

- **Chimney and spark arrestor (line 8).** 150 mm single-wall stainless flue about 0.5 m and a spark arrestor cap with mesh of 6 mm or finer.
- **Header and seal fittings (line 9).** DN50 stainless pipe and caps, sockets, 114.3 and 42.2 pipe, sight tube, drain valve; a three-way diverter valve rated for saturated steam that cannot stop with both outlets shut; a steam-rated quick coupling.
- **Relief valve (line 10).** Certified, code-stamped steam safety valve, 15 psi (1.03 bar) set, 3/4 in, rated 170 kg/h (375 lb/h) or more.
- **Steam hose (line 11).** EPDM saturated-steam hose, 25 bore, 6 m, ground-joint couplings and whip checks.
- **Battery (line 13), instruments (line 14), alarms (line 16).** As Figure 33; a 0 to 1 bar steam gauge with a siphon; four type K thermocouples and a reader.
- **Safety kit (line 15).** Personal carbon monoxide alarm, 2A:10B:C fire extinguisher, heat-resistant gloves, face shield.
- **Pipework and fittings (lines 19 and 20).** 3.5 m of 12.7 x 1.2 stainless tube for the feed line and inlet jumper; 1 m of 16 mm suction hose; stainless compression unions, a 25.4 to 12.7 reducer and two bulkhead unions; 3/4 in pipe and elbow for the relief discharge; 1 in pipe, two elbows and a tee for the diverter vent line.
- **Fixings and straps (line 21).** 4 x M12 x 160 bolts; 12 x M10 bolts and coach bolts; 14 x M8 x 25 bolts; 12 x M8 and 8 x M6 weld studs; 8 x M6 bolts; two M8 U-bolts for 60 mm pipe; nyloc nuts; two 25 mm ratchet straps with lashing points; 4 m of ceramic rope gasket.

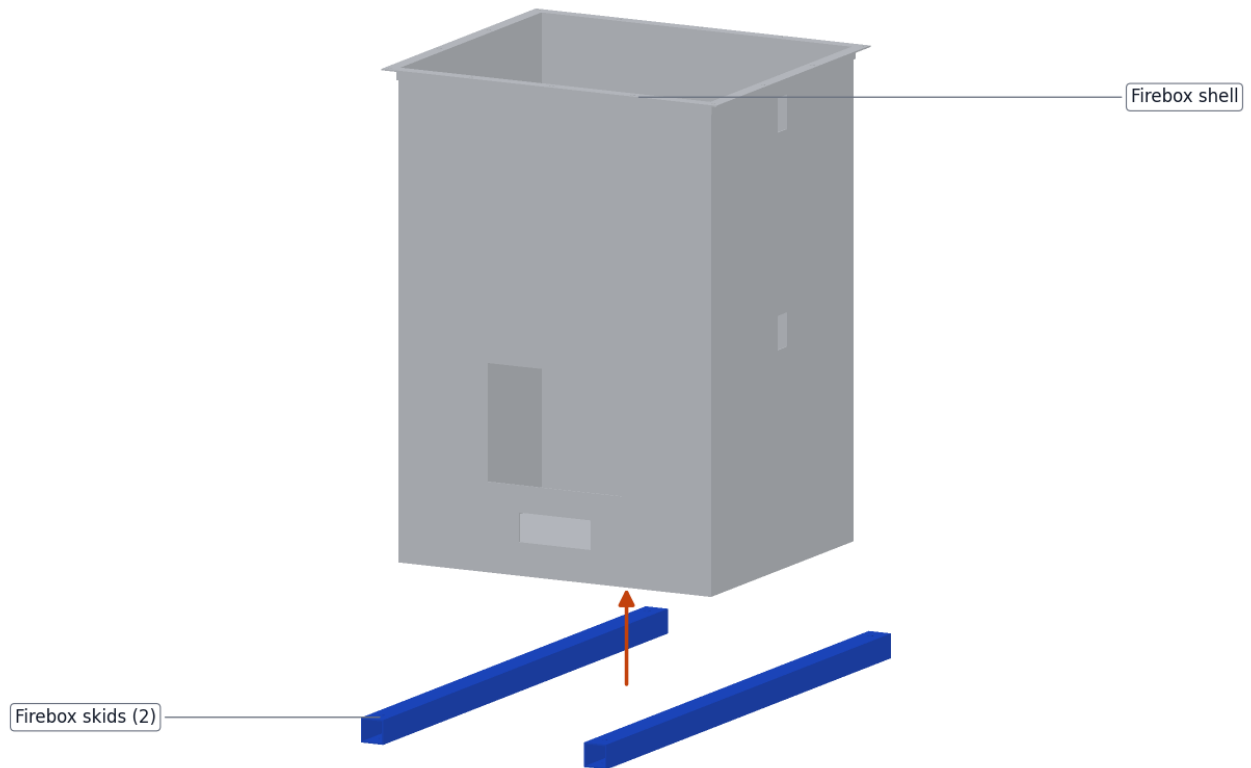
■ 4. Putting it together

In each picture the parts already fitted are grey and the part being fitted is in colour, with an arrow showing the way it goes in.

Step 1: skids under the firebox shell

Step 1: skids under the firebox shell

Shell standing on the bench. Two skids across the floor, 500 apart, stitch welded both sides



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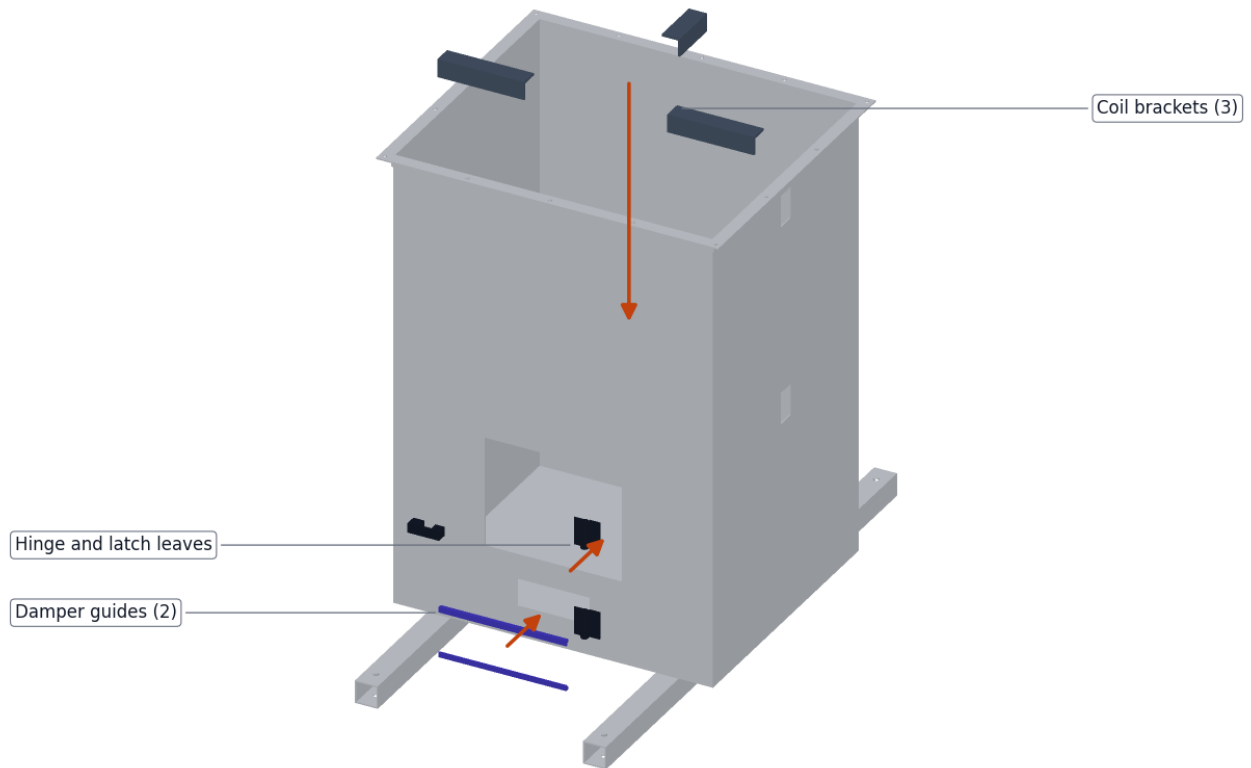
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With the shell upside down on the bench, lay the two skids across the floor, 500 apart and 100 in from each end, and stitch weld them both sides (section 3.3).

Step 2: weld on the brackets, guides and hinge leaves

Step 2: weld on the brackets, guides and hinge leaves

Coil brackets inside (stainless rod); damper guides, hinges, latch keeper, gland studs and clip lugs outside



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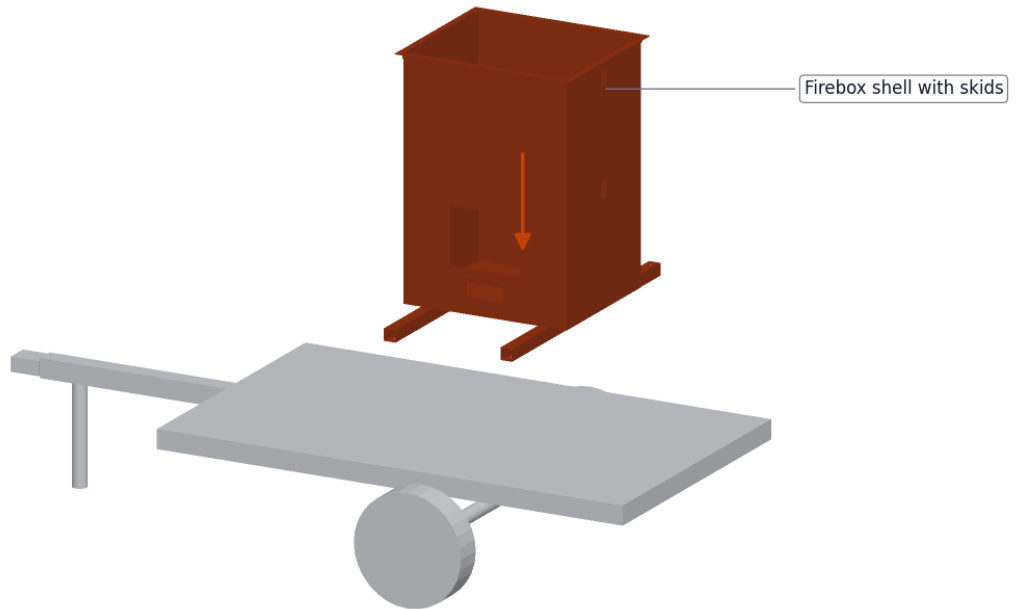
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Weld the three coil brackets inside (section 3.4) and the damper guides, hinge leaves, latch keeper, gland studs and clip lugs outside (sections 3.2, 3.7, 3.8). **Hold point:** all welding on the shell is finished before the lining goes in.

Step 3: firebox onto the trailer

Step 3: firebox onto the trailer

Lift with a hoist or four people. Skids across the deck; four M12 bolts through the skids, deck and side rails



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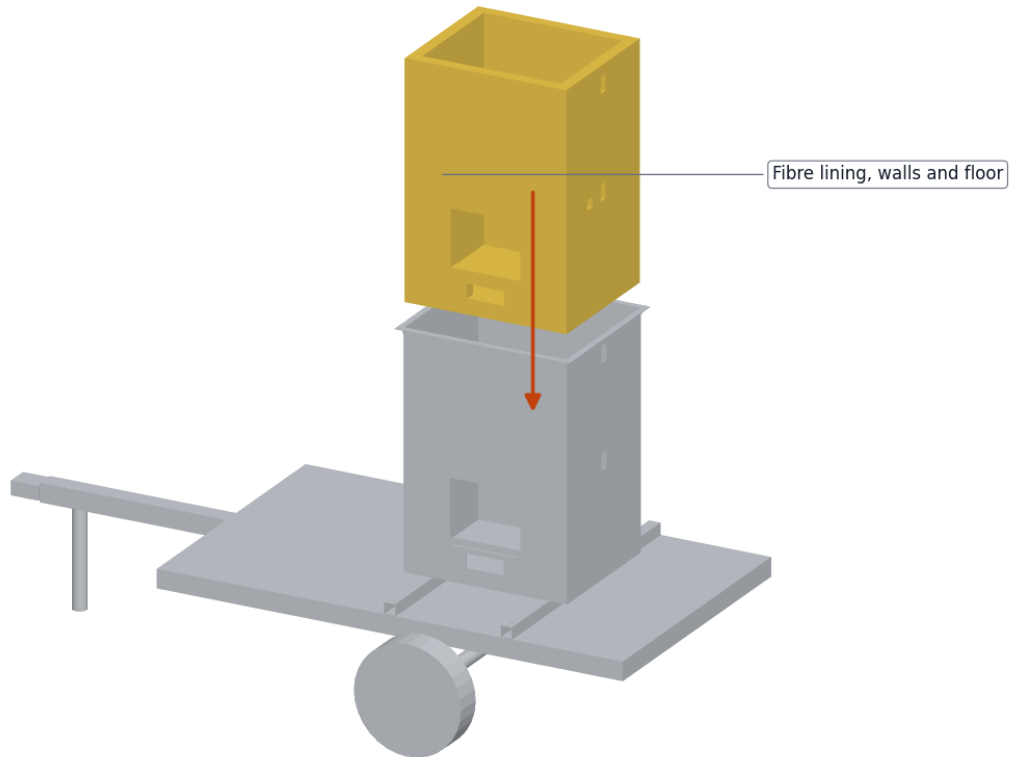
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Lift the shell onto the deck with a hoist or four people, skids across the deck, the firebox centred on the deck's centre line with its right face 400 from the right end of the deck. Drill through the skid holes into the deck and side rails and fit four M12 bolts, nyloc nuts under the rails.

Step 4: fibre lining into the firebox

Step 4: fibre lining into the firebox

Floor board first, then two layers of blanket on the walls; cut round the door, air inlet, slots and brackets



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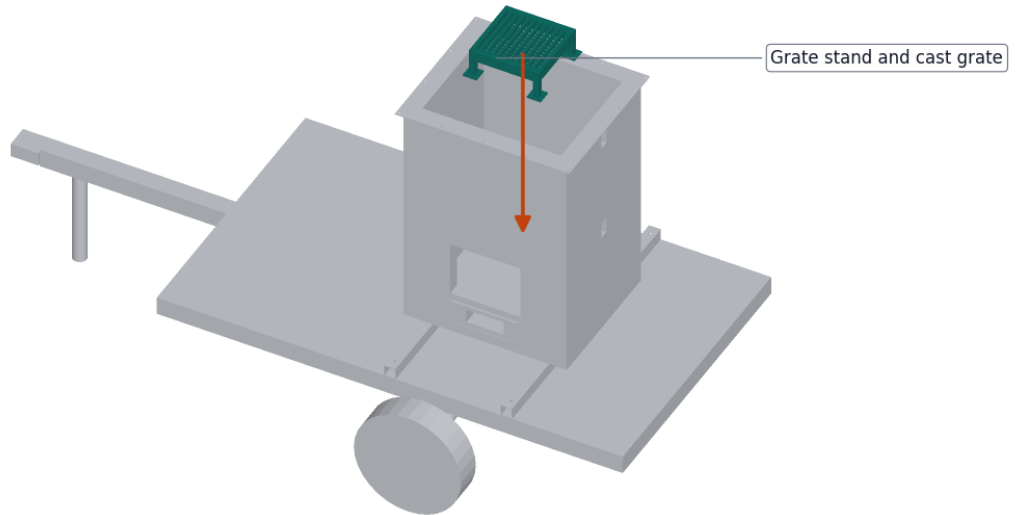
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Floor board first, then the two layers of wall blanket over the pins, cut round the openings, slots and brackets (section 3.5). Wear the protective equipment in section 7.

Step 5: grate stand and grate

Step 5: grate stand and grate

Lowered in through the open top; stand in the middle of the floor, grate loose on top



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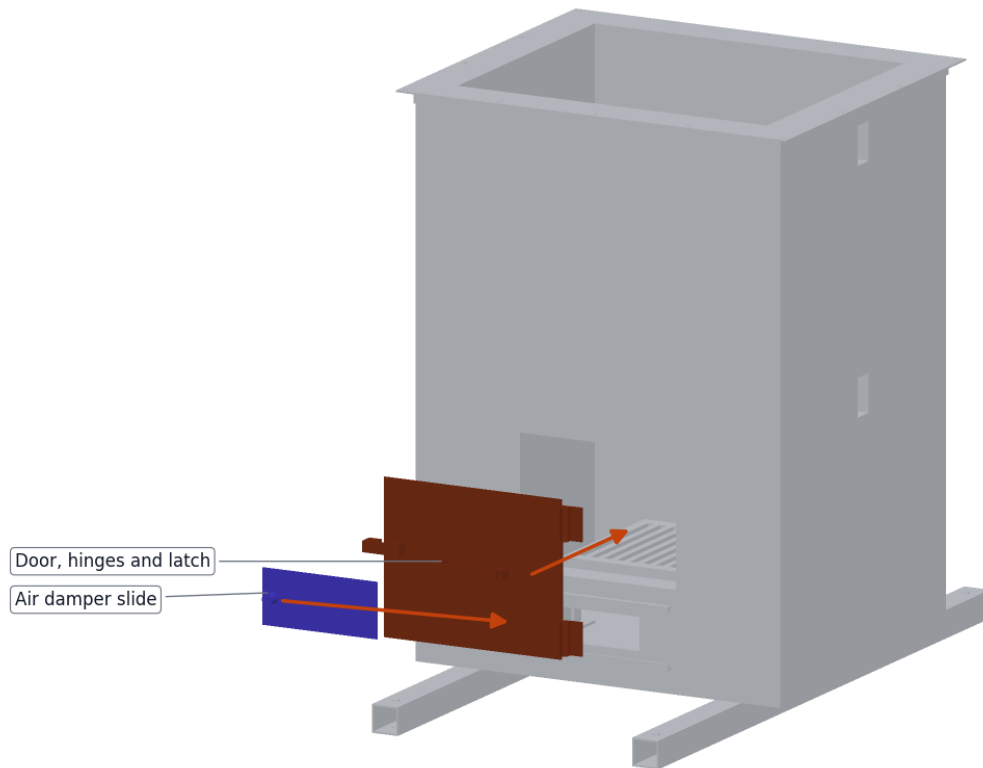
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Lower the stand in through the open top onto the middle of the floor board and set the grate on it.

Step 6: door and air damper

Step 6: door and air damper

Hang the door on its lift-off hinges; glue rope round the opening. Slide the damper into its guides from the side



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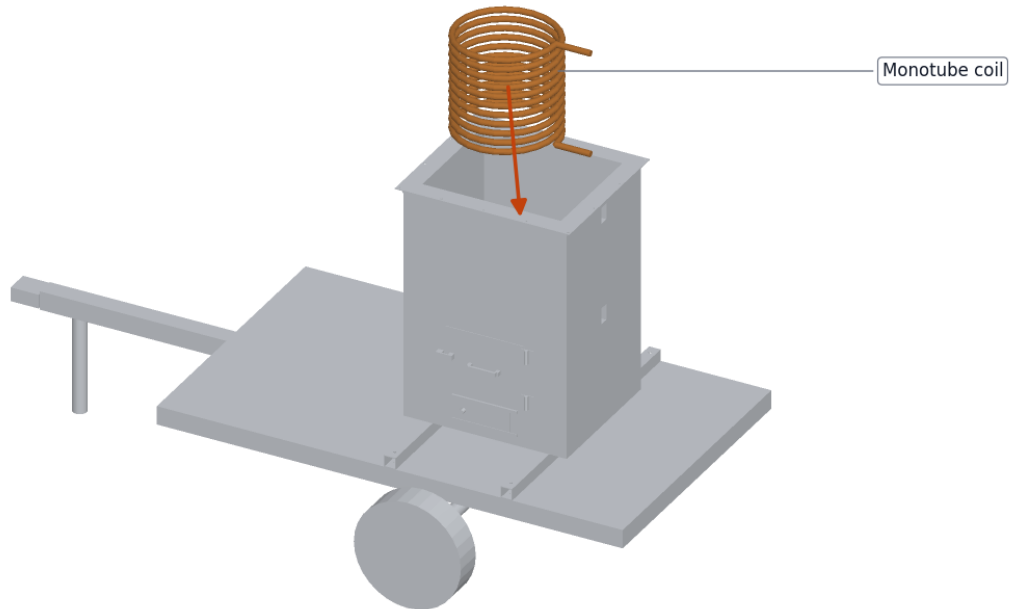
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Lift the door onto its hinges and check it latches. Slide the damper into its guides from the side.

Step 7: coil into the firebox

Step 7: coil into the firebox

Lower it in 66 mm toward the left so the tails clear; at 30 mm above the brackets slide it right, tails out through the slots; set it down



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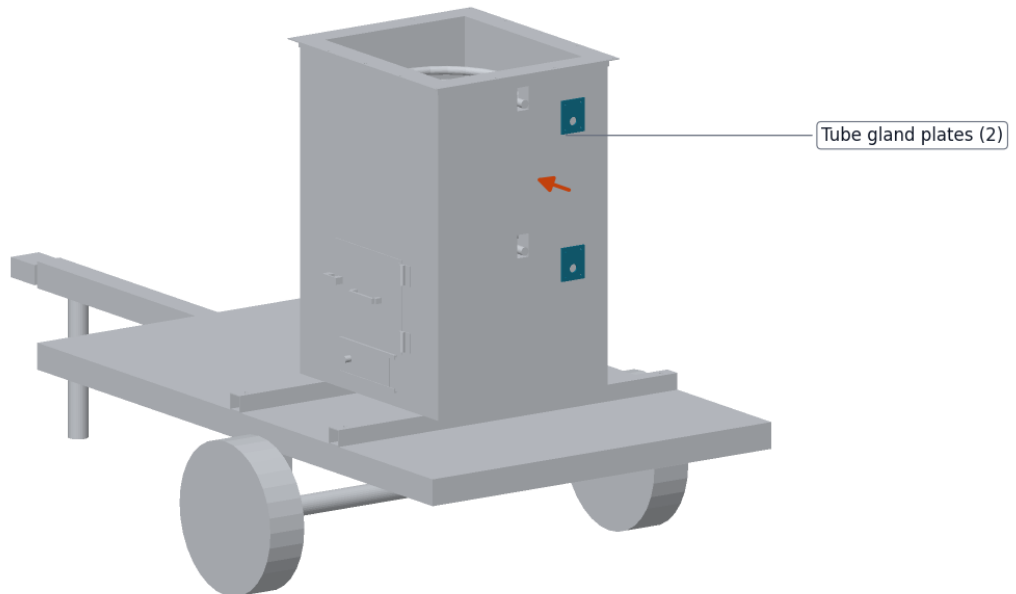
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With a helper, lower the coil in through the open top with its tails toward the right end, held 66 toward the left and 30 above the brackets, so the tail ends clear the lining. Slide it to the right so the tails pass out through the slots, then set it down onto the three brackets. **Hold point:** the coil rests on all three brackets and both tails sit in the bottom of their slots; shim a bracket if needed.

Step 8: gland plates over the coil tails

Step 8: gland plates over the coil tails

Pack each slot round the tail with ceramic rope, slide the plate on, nip the four M6 nuts. Tube must still slide



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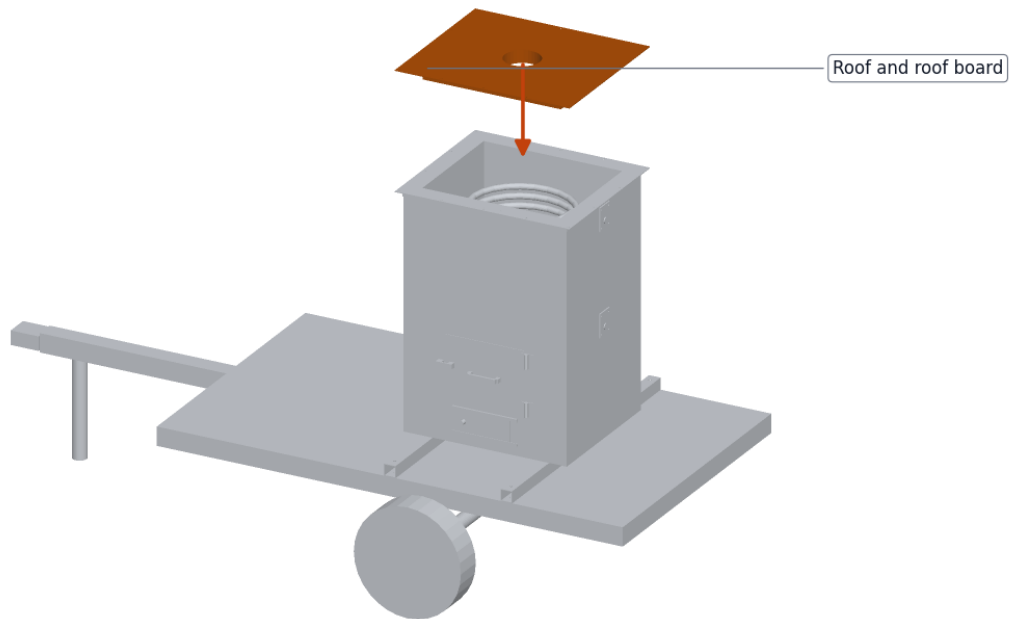
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Pack each slot round its tail with ceramic rope, slide a gland plate over each tail end and nip the four M6 nuts so the rope seals but the tube still slides.

Step 9: roof onto the firebox

Step 9: roof onto the firebox

Ceramic rope gasket on the top frame; roof with its board lowered on; 14 M8 bolts round the edge



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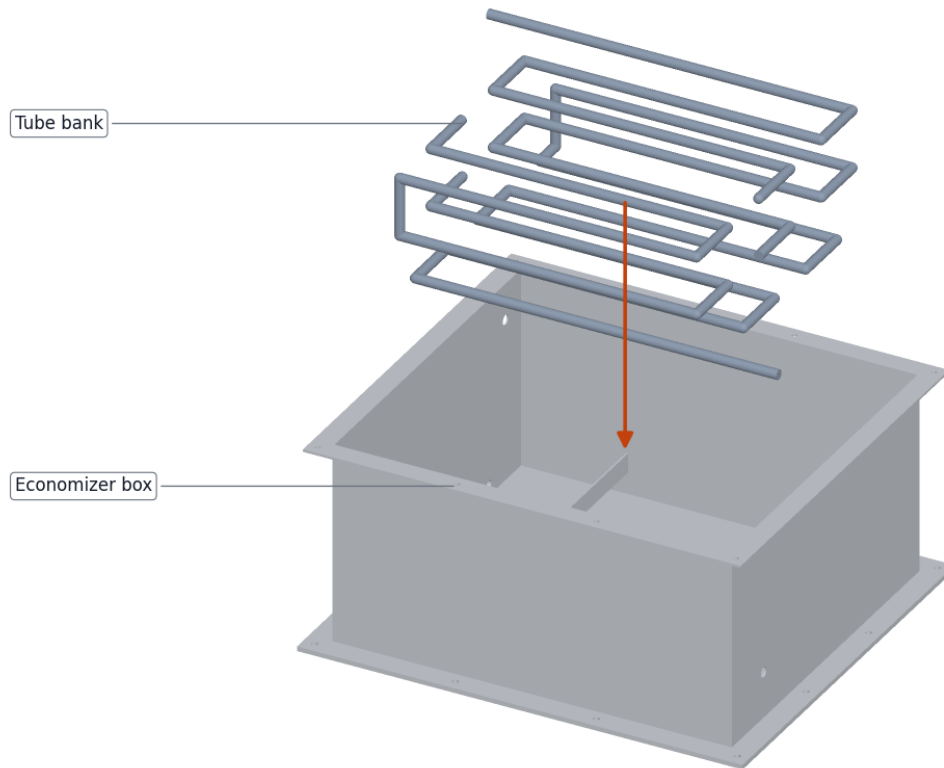
github.com/BoujeeEnjinia1701/steamroot

Lay ceramic rope on the top frame, lower the roof and board on, and fit the 14 M8 bolts, nuts under the frame, tightened evenly.

Step 10: tube bank into the economizer box (on the bench)

Step 10: tube bank into the economizer box (on the bench)

Lower the bank onto the two support bars; connect each end to its bulkhead union through the end wall



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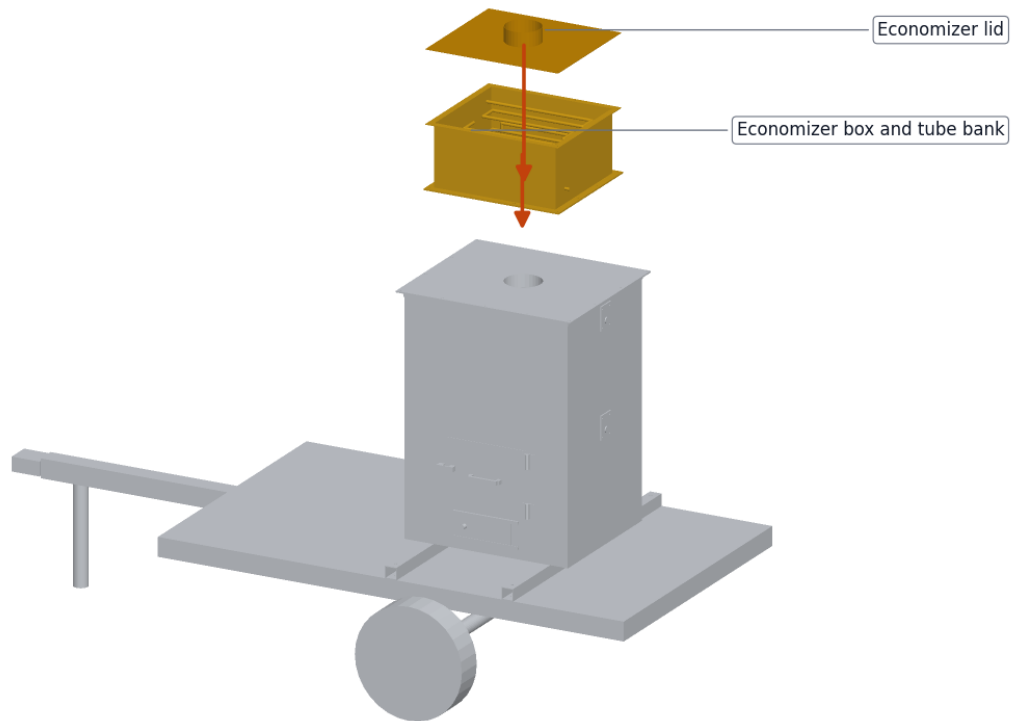
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Lower the tube bank onto the two support bars and connect each end to its bulkhead union.

Step 11: economizer and lid onto the roof

Step 11: economizer and lid onto the roof

Rope gasket on the roof; box over the twelve welded studs, nuts on top. Lid on its gasket, eight M6 bolts



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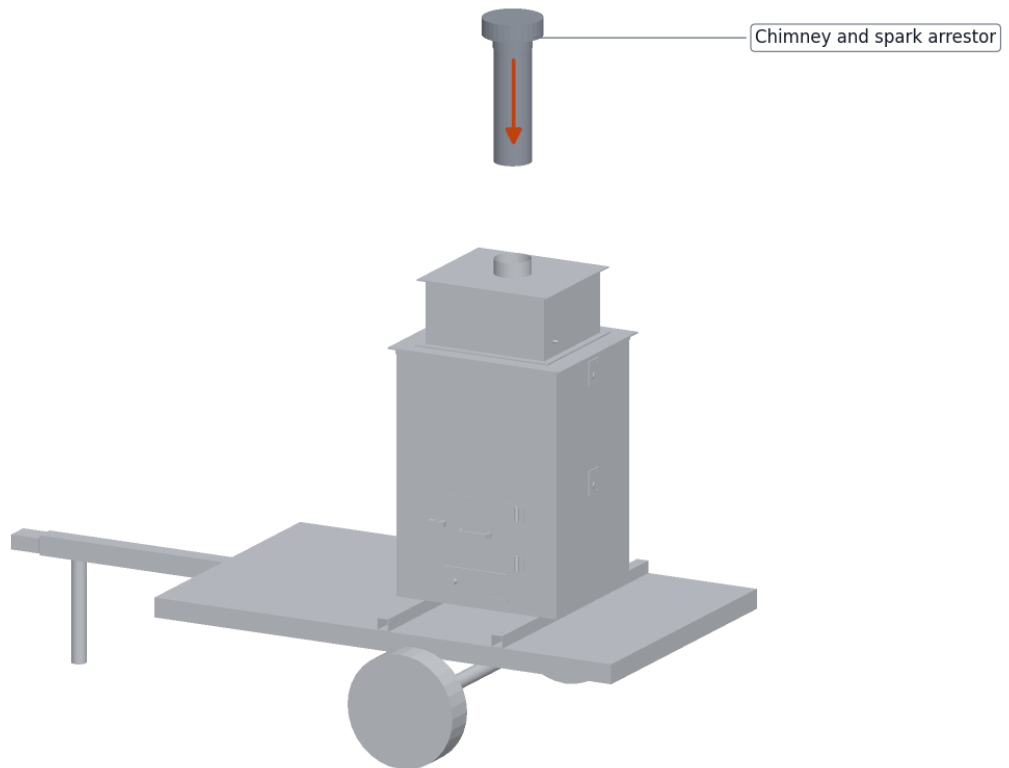
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Lay rope gasket on the roof round the flue hole, set the economizer over the twelve studs and fit the nuts. Lay rope on the top flange and bolt the lid on with eight M6 bolts.

Step 12: chimney and spark arrestor

Step 12: chimney and spark arrestor

Chimney slid over the spigot, three self-drilling screws; cap pushed on top, mesh in place



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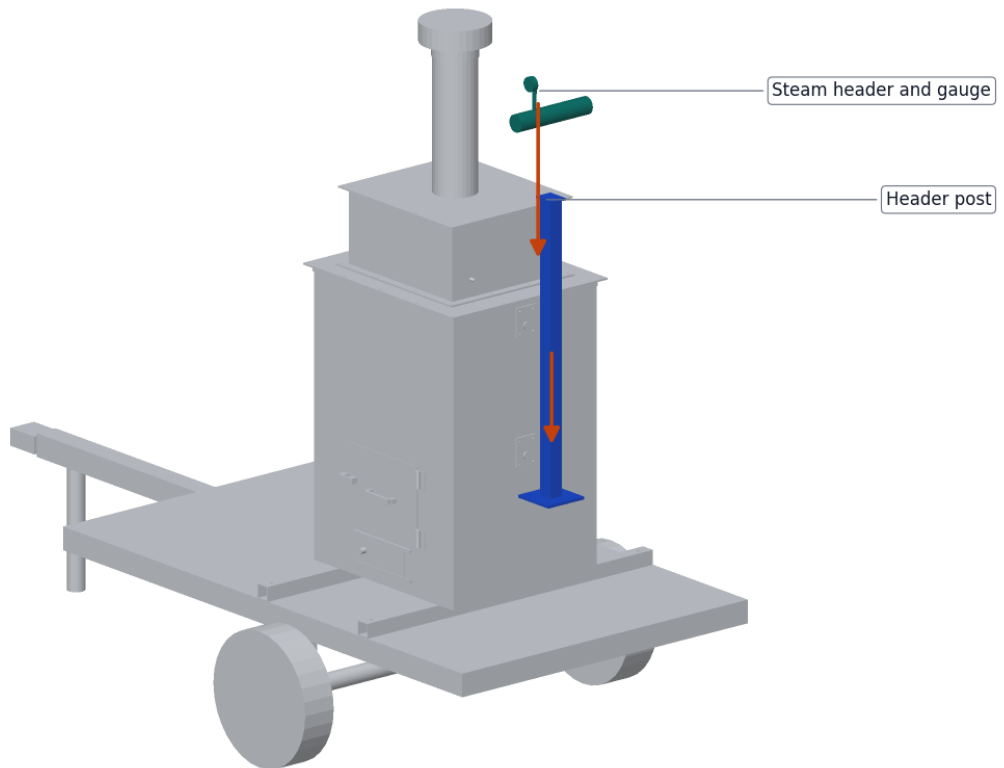
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Slide the chimney over the spigot and fix it with three self-drilling screws; push the spark arrestor cap onto its top. Check the mesh is in place.

Step 13: header post and header

Step 13: header post and header

Post on the deck, four M10 bolts. Header on the saddle under two U-bolts, sockets as the header sketch shows



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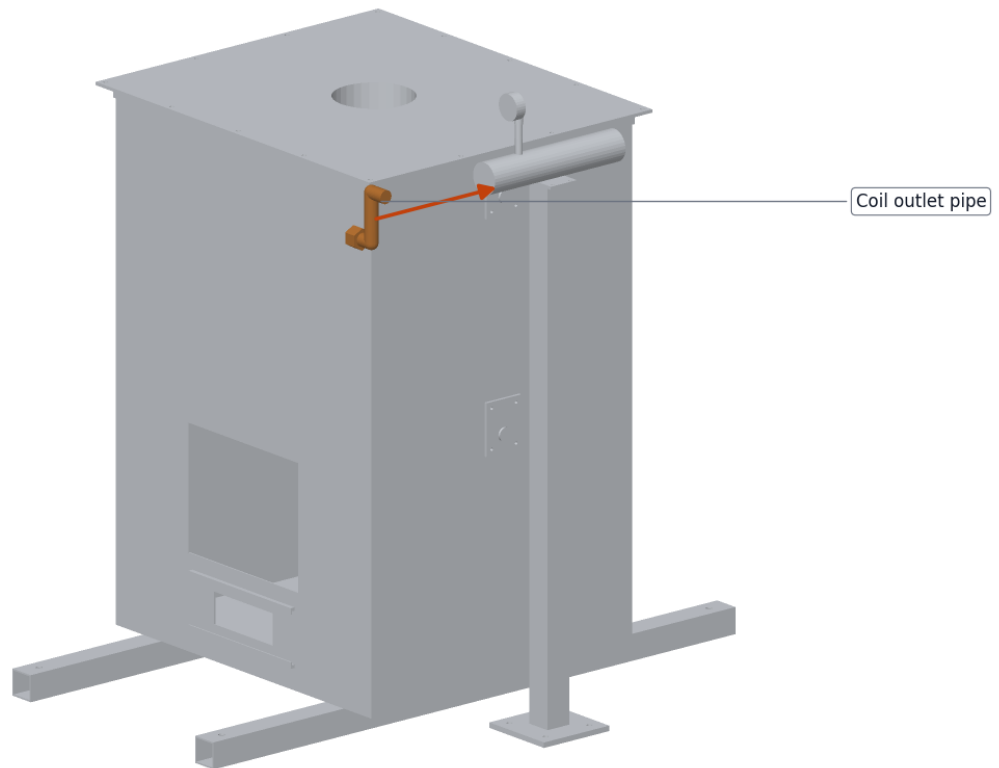
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Stand the post on the deck 130 right of the firebox on the centre line, mark and drill the deck, and fit four M10 bolts. Lay the header on the saddle, sockets as section 3.16, and fit the two U-bolts.

Step 14: coil outlet pipe to the header

Step 14: coil outlet pipe to the header

Union onto the upper coil tail, other end into the header's side socket. Tighten the union last, without strain



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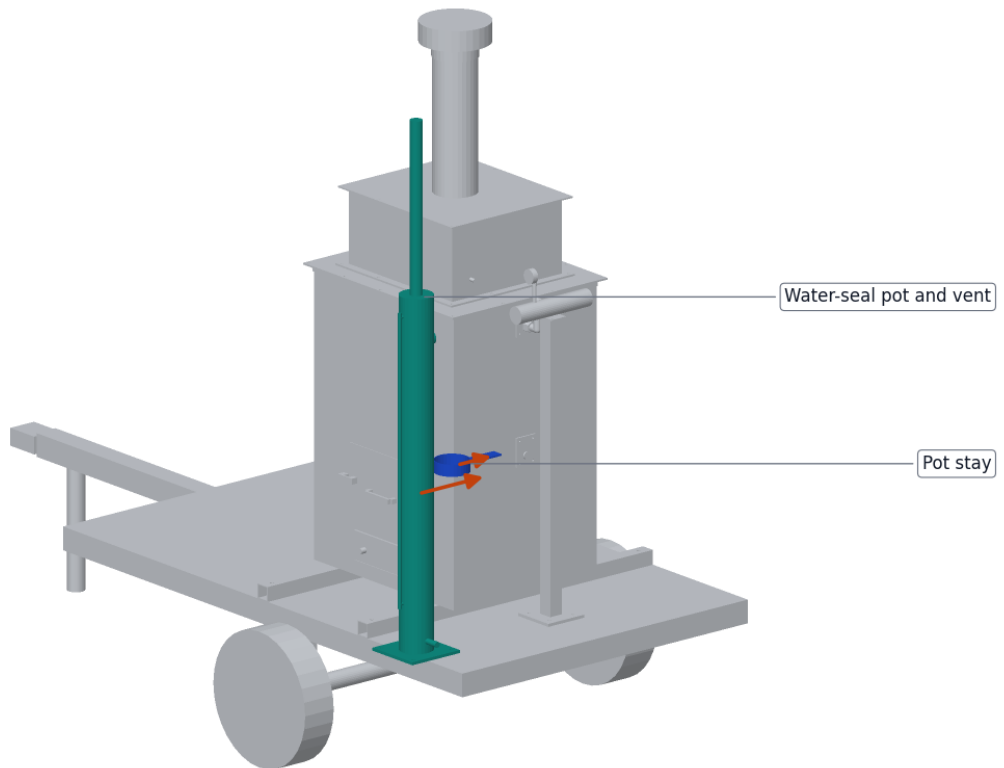
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Bend the outlet pipe from 25.4 stainless tube to fit from the union on the upper tail to the header's side socket: out 25, up 94, across 40. Fit the union on the tail and screw the other end into the header. Tighten the union last, so nothing strains the coil.

Step 15: water-seal pot and stay

Step 15: water-seal pot and stay

Pot on its foot plate, four M10 bolts; branch into the header's front cap; stay band round the pot



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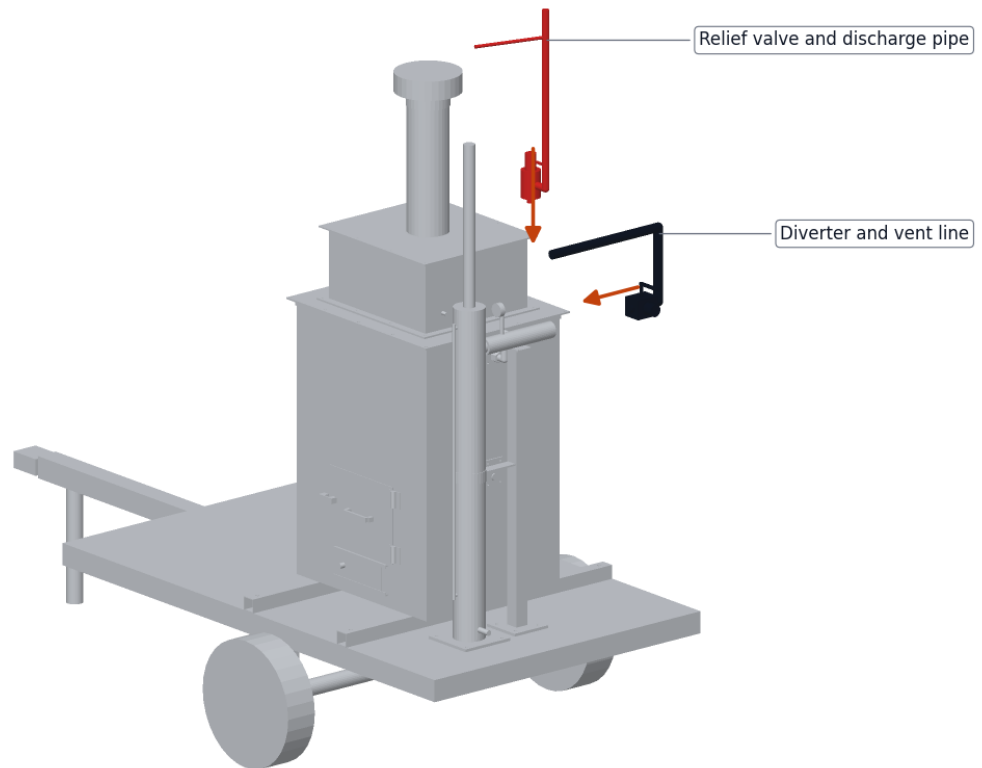
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Stand the pot on the deck in front of the post, its branch lined up with the header's front cap. Mark, drill and bolt its foot plate (four M10). Join the branch; bolt the stay's band round the pot and weld the stay to the post.

Step 16: relief valve and diverter

Step 16: relief valve and diverter

Relief valve upright on top of the header, discharge pipe to 2.3 m, stay to the vent. Diverter on the back nipple, vent line into the vent



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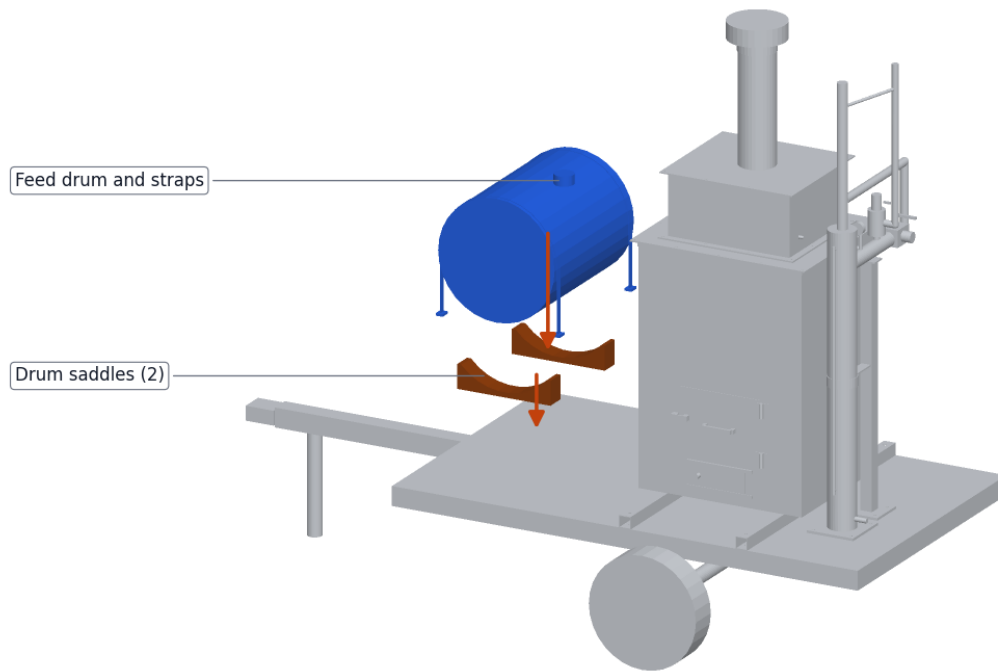
github.com/BoujeeEnjinia1701/steamroot

Screw the relief valve upright into its socket on top of the header, lever toward the front, and fit the discharge pipe from its outlet up to 2.3 m with the stay to the vent pipe. Screw the diverter onto the header's back nipple with its hose coupling pointing right, and fit the vent line from its vent port up and into the vent pipe's tee. **Hold point:** safety stop S4.

Step 17: feed drum on its saddles

Step 17: feed drum on its saddles

Saddles across the deck, two M10 coach bolts each. Drum in the saddles, outlet toward the pump box; two ratchet straps



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

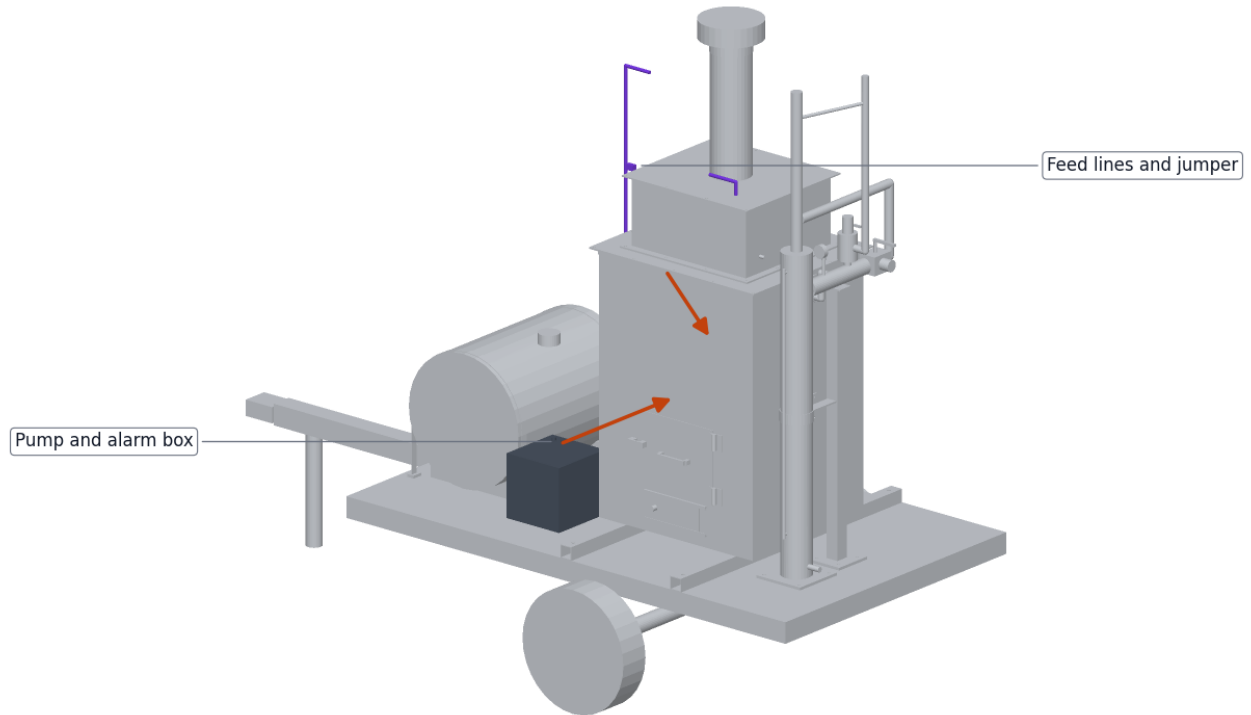
github.com/BoujeeEnjinia1701/steamroot

Bolt the saddles across the deck, 440 apart, with M10 coach bolts. Lay the drum in them with its outlet toward the back, and fit the two ratchet straps to the lashing points.

Step 18: pump box and feed lines

Step 18: pump box and feed lines

Box bolted to the deck. Suction hose drum to pump; stainless line pump to economizer inlet, in two clips; jumper to the coil inlet



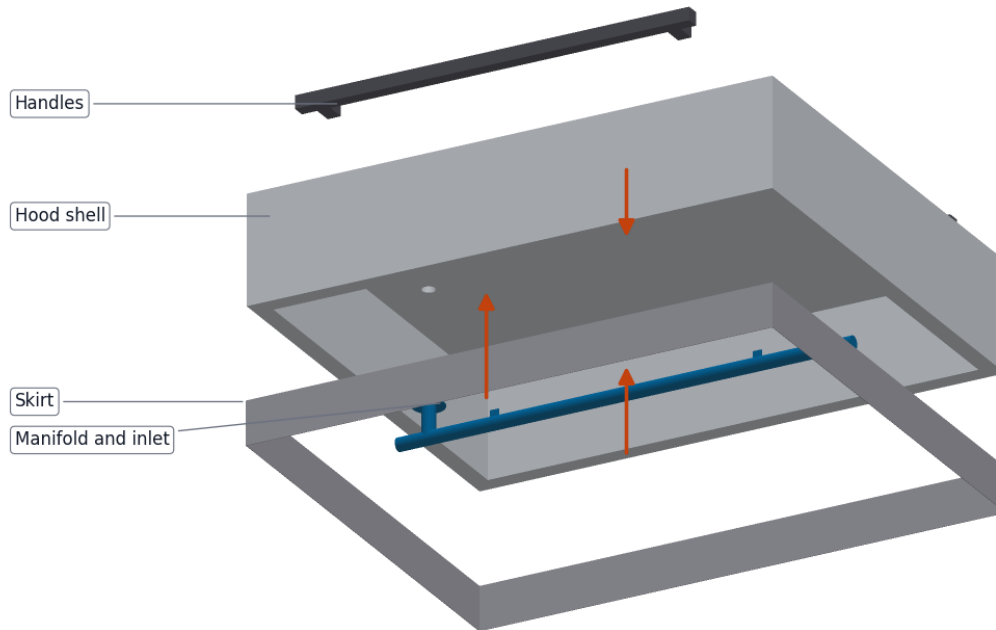
BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/steamroot

Bolt the pump and alarm box to the deck at the back edge, just right of the drum. Run the suction hose from the drum outlet to the pump. Bend the stainless feed line from the pump up the left end of the firebox, through its two clips, to the economizer inlet union. Fit the reducer and union on the lower coil tail and bend the jumper up the right end to the economizer outlet union. **Hold point:** safety stop S3.

Step 19: build each hood (on the bench, make two)**Step 19: build each hood (on the bench, make two)**

Hood upside down: hang the manifold, riser up through the top; skirt riveted round the outside; handles on their blocks



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

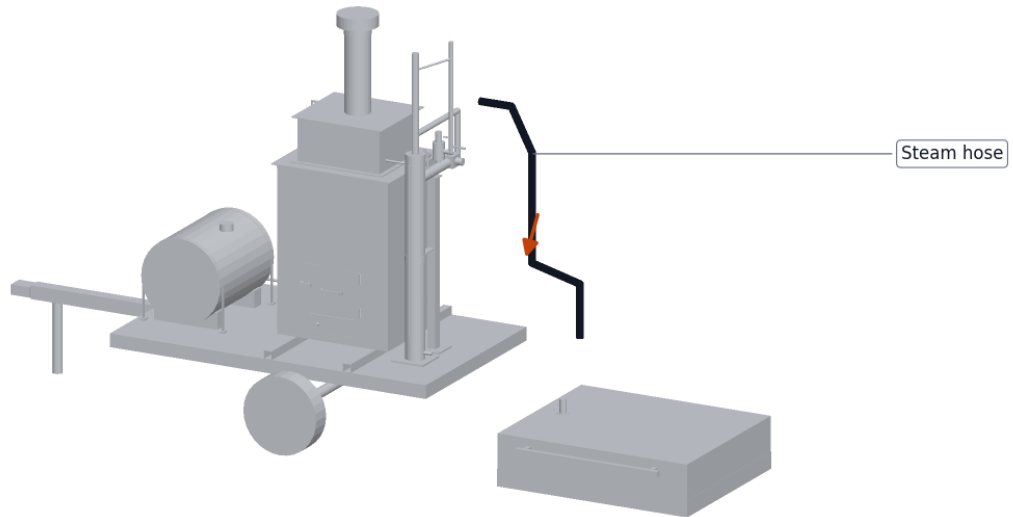
github.com/BoujeeEnjinia1701/steamroot

With the hood upside down, hang the manifold from the inner skin on its two straps with the riser up through the inlet hole; bolt the flange down on sealant on top. Rivet the skirt round the outside. Bolt the handles to their standoffs through the hardwood blocks.

Step 20: steam hose to a hood

Step 20: steam hose to a hood

Hose from the diverter coupling to the hood inlet, whip checks fitted at both ends. Moved only with steam to the vent



BUILD PLAN ILLUSTRATION, PLAN NOT YET BUILT

github.com/BoujeeEnjinia1701/steamroot

Couple the hose to the diverter and to the hood in use, and fit the whip checks at both ends. **Hold point:** safety stops S5 and S6.

■ 5. First checks

These are the checks a TRL 4 test report would record; this plan only lists them. Requirement numbers are those of STR-REQ-001.

Table 2. First checks.

Check	Requirement	How	Pass when
Open path to the vent	R6	Cold, empty: with the diverter in each of its positions, blow low-pressure air into the coil inlet	Air always leaves by the vent or the hose; no position closes both
Seal limit	R6	Pot filled to the mark with cold water; raise the header pressure slowly with air through a regulator; watch the gauge	Steam path opens (air bubbles through) at 0.09 to 0.10 bar; no water leaves the vent
Water in the heated section	R7	Fill the coil, economizer and header; drain and measure	8 L or less (7.0 L calculated)
Leak test, steam side	R6, R7	Cold water at 3 bar for 10 minutes, seal pot branch plugged for the test only	No leak, no pressure fall; plug removed
Feed rate	R3, R8	Set the needle valve; time a measured volume at the coil inlet	0.5 L/min (30 kg/h)
Alarms	Feed control (STR-DDR-001)	Heat the thermocouple with a hot air gun; lift the float switch	Alarm sounds at about 150 °C and at about 20 L left
Relief valve	R9	Read the nameplate and code stamp; lift the test lever cold	15 psi set; 170 kg/h or more; lever moves freely
Chimney and mesh	R10	Measure the outlet height and the mesh	2.2 m or more (2.4 m); mesh 6 mm or finer
Mass and width	R11	Weigh on a weighbridge with the drum and pot drained; measure the width	500 kg or less (468 kg estimated); 1.5 m or less (1.48 m)
Steam output and pressure (first fire)	R3, R6	With feed at 0.5 L/min, fire to steady steam at the hood; read the header gauge	Dry steam at the hood; header 0.1 bar or less
Hood handles	R10	Contact thermometer on the handles and outer skin after 30 minutes of steaming	60 °C or less
Run per fill	R8	Time from a full drum to the low-level alarm	3 h or more (4.2 h calculated)
Soil temperature	R1	Type K probes at 15 cm under a hood	70 °C or more for 20 to 30 minutes over 80 % of the footprint

■ 6. Safety stops

Stop at each point. Carry on only when everything listed is true.

- **S1. Before anything is bought or built.** A written ruling from the local boiler authority on this open-vented, wood-fired monotube coil is in hand, and the design has been checked against it. Its ruling overrides this plan where it is stricter.
- **S2. Before welding, grinding or cutting fibre.** Hot work area clear of anything that burns, extinguisher at hand, welding screen up. Cut ceramic fibre with a knife, damp, never with a power saw; wear a P2 (FFP2) mask, gloves and long sleeves, and vacuum, never sweep. Never weld galvanised steel.
- **S3. Before any water goes into the coil or economizer.** Every joint is made up; the coil and economizer have passed their 3 bar cold tests; the seal pot branch is open (no test plug left in).

- **S4. Before the seal pot is put in service.** The vent is open from the pot to its outlet at 2.3 m; the diverter cannot stop with both outlets shut; nothing on the steam side can isolate the coil from the vent; the relief valve is fitted upright with its discharge pipe open to 2.3 m; the pot is filled to the mark.
- **S5. Before the first fire.** All first checks up to "Run per fill" in section 5 pass, except the ones that need fire. The machine stands on bare ground or gravel, clear of dry vegetation, outdoors, with no fire ban and no high wind. Extinguisher and water at hand; the operator wears a personal carbon monoxide alarm, heat-resistant gloves, a face shield, long sleeves and closed boots; bystanders, children and animals at least 5 m away. The feed pump is running and water flows before the fire is lit: never fire a dry coil.
- **S6. Before steam goes to a hood.** The diverter is set to the vent; the hose and whip checks are inspected; the hood is set on the soil with its skirt pressed in. Keep hands and feet clear of the skirt and the vent outlet.
- **S7. Before lifting a hood or moving the hose.** Steam has been diverted to the vent for at least one minute.
- **S8. After any alarm.** Close the air damper. Never restart the feed into a hot, dry coil: let the fire die and the coil cool first.
- **S9. Before towing.** The fire is out and cold; the drum and seal pot are drained; the hoods are off the trailer; the lights work.

■ 7. Tools, skills and workspace

Tools. MIG welder for 2 to 3 mm mild steel; stick or TIG welder with 309 rod for the stainless brackets and 316 rod for the stainless header; angle grinder with cutting, grinding and flap discs; plasma cutter or jigsaw with metal blades for sheet; bench drill and drills to 13 mm; step drill; 150 mm hole saw or plasma for the flue holes; sheet metal folder (or clamped angle and mallet) for the 1 mm hoods and the 1.5 mm skirts; pop rivet tool; hand tube bender for 1/2 in tube, 38 mm radius; a 395 mm coil former and spacer blocks; pipe wrenches, spanners and a torque wrench; thread sealant rated for steam; utility knife for the fibre; tape measure, square, level and spirit level; hydraulic hand pump and gauge to 4 bar for the cold tests; multimeter; hot air gun; a hoist or four people for lifting the firebox.

Skills. Welding mild steel sheet and tube to a sound, leak-free standard, and stainless to mild steel; bending tube without kinks; threaded and compression pipe fitting; folding and riveting sheet; 12 V wiring. Steam pipework must be fitted by someone who has done it before or is supervised by someone who has. No mains wiring is part of this build.

Workspace. A fabrication workshop with a concrete floor and room for the trailer and a 2 x 1 m bench; a separate bench for the hoods; good ventilation for welding; a wet area for the cold pressure tests. First fire outdoors only, on bare ground or gravel.

Personal protective equipment. Welding helmet, gloves and jacket; safety glasses and hearing protection for grinding and cutting; P2 (FFP2) mask, gloves and long sleeves for ceramic fibre and mineral wool; heat-resistant gloves, face shield, long sleeves and closed boots for any work near a fire or steam; a personal carbon monoxide alarm at any fire.

■ 8. Where the numbers come from

- Model and constructability checks: `cad/src/model.py` (`python cad/src/model.py --check` , 93 checks); STEP and STL exports in `cad/step/` and `cad/stl/` .
- Pictures: `cad/src/build_plan_media.py` , using `.kit/build_views.py` ; written to `docs/05-build-plan/` and `cad/drawings/STR-DWG-101` to `STR-DWG-121` .

- General arrangement: `cad/drawings/STR-DWG-002.pdf` , Rev P3.
- Calculations: `docs/04-calcs/01-sizing.md` (STR-CAL-001 v0.3) and `docs/04-calcs/sizing.py` ; masses section 7, pressures and seal section 5, stored water section 6.
- Bill of materials: `bom/bom.csv` .
- Decisions: `docs/decisions/0003-design-for-construction.md` (STR-DDR-003), with STR-DDR-001 and STR-DDR-002; open decisions in `docs/06-design-decisions.md` (STR-DEC-001).
- Requirements: `docs/03-requirements.md` (STR-REQ-001 v0.5).